
Relaxation and the NOE

In this chapter we are going to look at the phenomenon of *relaxation*, which is how the bulk magnetization from the spins reaches its equilibrium value. As we have already seen, at equilibrium we have magnetization along z , but none in the transverse plane: it is relaxation which drives the spins to this equilibrium state.

This chapter starts out by describing how relaxation comes about, and how its rate is determined by molecular properties, such as shape and motion. We then move on to consider some important applications of relaxation phenomena, such as the nuclear Overhauser effect (NOE) and cross correlation.

Relaxation in NMR is unusually slow when compared with that of other molecular energy levels. For example, the lifetime of an excited electronic state is typically a few microseconds, and the vibrational and rotational energies of molecules are likely to change at every collision, giving them lifetimes of a few nanoseconds. In contrast, it takes between milliseconds and seconds for the equilibrium magnetization to be established; in extreme cases this process can take minutes.

The slow relaxation is at once both an advantage and a disadvantage. The advantage of slow relaxation is that it means that any transverse magnetization (or, more generally, coherences) we generate will survive for long enough for it to be manipulated and observed. Multiple-pulse NMR, with its sequences of pulses and delays lasting many milliseconds, would simply not be possible if relaxation were not so relatively slow. Furthermore, slow relaxation means that the FID persists for long enough for us to obtain high-resolution spectra – that is, the linewidths are narrow.

The disadvantage of slow relaxation is that it sets a lower limit on the rate at which an experiment can be repeated. Remember that, in even the simplest pulse–acquire experiment, we usually repeat the experiment several times in order to improve the signal-to-noise ratio. The problem is that, before we can repeat the experiment, we have to allow sufficient time for the equilibrium magnetization to be re-established. If we do not do this, the z -magnetization will be reduced in size, and so will the observed signal.

We will see in this chapter that in NMR the rate of relaxation is sensitive to the physical environment the nuclei find themselves in and the nature of the motion which the molecule is undergoing. Thus relaxation can be used as a probe for studying both of these things. It turns out that, when compared with other techniques, NMR is sensitive to rather slow molecular motions, and so NMR can be a unique source of such information.

When it comes to structural studies, the one really important manifestation of relaxation is the NOE. This allows us to identify which nuclei are close in space (for protons, say within 5 to 6 Å), thus providing complementary information to scalar coupling, which relates to the bonding network. Although the NOE is only semi-quantitative when it comes to estimating distances, it nevertheless forms a key element in the NMR spectroscopist's arsenal of techniques. Much of this chapter will therefore be devoted to understanding the origin of the NOE, and to describing the experiments which are used to measure it.

We will start with a description of what relaxation is, focusing on how what is happening to the individual spins leads to a particular behaviour of the bulk magnetization of the sample. Our discussion will reveal the importance of the timescale of molecular motions, and introduce the key idea of the *correlation time*, which is used to characterize these motions.

Having established the underlying physical phenomena which are responsible for relaxation, we will go on to describe the origin of the NOE. To do this we will need to consider in detail the relaxation properties of a pair of spins, something which is conveniently described using the *Solomon equations*. These equations are also a convenient way of understanding how the experiments used to detect the NOE work.

Having discussed the NOE, we then go on to look at the relaxation of transverse magnetization; this kind of relaxation has some similarities and some important differences to the relaxation of z -magnetization. The chapter closes with a discussion of the phenomena of *cross correlation* which results from the interference between different sources of relaxation.

Describing the effects of relaxation is relatively straightforward, but understanding the origin of relaxation and its dependence on motion is rather more difficult. It is probably the part of NMR theory that most people, including the author, find most difficult to grasp. Do not be surprised, therefore, if you find this chapter more challenging than those which have preceded it.

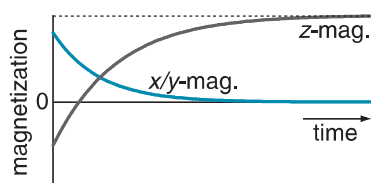


Fig. 9.1 Relaxation drives the z -magnetization to its equilibrium value, indicated here by the dashed line, and the transverse (x or y) magnetization to its equilibrium value of zero.

9.1 The origin of relaxation

Relaxation is the process by which, over time, the bulk magnetization returns to its equilibrium position. We have already established that at equilibrium there is *no* transverse magnetization, but there is z -magnetization along the direction of the applied magnetic field. The size of this equilibrium z -magnetization depends on the number of spins, their gyromagnetic ratio and the size of the applied field.

Thus, relaxation drives the transverse magnetization to zero and the longitudinal (that is, along z) magnetization to a particular steady value, as illustrated in Fig. 9.1. The description of the equilibrium magnetization

as being ‘steady’ is important, as by definition at equilibrium all quantities cease to be time dependent i.e. nothing is changing.

However, this description tells us nothing about what is happening to the individual spins which results in this behaviour of the bulk magnetization. How the two are connected is the topic of the next few sections.

9.1.1 Behaviour of individual magnetic moments

It is helpful at this point to remind ourselves how the bulk magnetization of the sample is related to the magnetic moments of individual spins. Recall the discussion in section 4.1 on page 47, where we described how each spin has associated with it a magnetic moment. This moment is a vector quantity, having a particular magnitude and direction. The magnitude is determined by the gyromagnetic ratio of the nucleus, but there are no restrictions on the direction in which a particular moment can point, so in general each moment has an x -, y - and z -component.

The bulk z -magnetization is found by adding together the z -components of the magnetic moment from each spin. Similarly, the x - and y -magnetizations are found by adding together the corresponding components from each spin. At equilibrium, the x - and y -components of the individual spins are distributed randomly, so adding them up results in no bulk transverse magnetization.

For the z -magnetization the story is rather different. As was described in section 4.1 on page 47, there is an energetic preference for the z -component of the magnetic moment to be aligned along the field direction. However, the energy preference is rather small and so the alignment is easily disrupted by the thermal motion. Nevertheless, when summed over the whole sample there is a net magnetization along the z -direction.

When an RF pulse is applied to equilibrium magnetization, the z -magnetization is rotated towards the transverse plane. As a result, the z -component is reduced in size. After the pulse, the resulting transverse magnetization precesses about the field direction, and it is this precession which we detect in the form of the FID.

The question is, what is happening to the magnetic moments of the individual spins during the pulse and period of free precession? It turns out that each magnetic moment behaves in *exactly* the same way as the bulk magnetization. That this is true can be seen from a quantum mechanical treatment of the motion of a single spin – the details of which are given in Chapter 6. However, even without looking into the details of the quantum mechanics, it seems reasonable that each magnetic moment behaves in the same way as the bulk magnetization, as the latter is composed of the former.

For example, if we apply a 90° pulse about the x -axis, the z -component of the magnetic moment is rotated to $-y$, the x -component is unaffected, and the y -component is rotated to $+z$. During free precession, the magnetic moment precesses about the z -axis at the Larmor frequency, sweeping out a cone of constant angle, as illustrated in Fig. 9.2. As a result we have a constant z -component, and oscillating x - and y -components.

At equilibrium, there is a net alignment of the individual moments along the z -axis. The effect of a $90^\circ(x)$ pulse is to rotate the z -component of each magnetic moment onto the $-y$ -axis. Therefore, after the pulse there is net

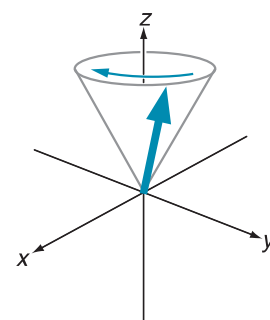


Fig. 9.2 Each spin has associated with it a magnetic moment, which is a vector quantity, having a magnitude and a direction. The moment can point in any direction, with its energy being determined by the angle between the moment and the applied magnetic field, which is along z . The moment precesses about the z -axis at the Larmor frequency, describing a cone of constant angle. As a result, the x - and y -components of the moment oscillate at the Larmor frequency. If a transverse magnetic field, oscillating at the Larmor frequency, is applied, the magnetic moment will be rotated in such a way that the angle it makes to the z -axis will be altered.

y-magnetization. The crucial thing is that the pulse affects each magnetic moment *in the same way*, so all the z-components are rotated onto -y. It is for this reason that the pulse is able to rotate the net z-magnetization into the transverse plane.

After the pulse, the magnetic moments of the individual spins precess about the z-axis at the Larmor frequency. Once again, the key thing is that each magnetic moment is precessing at the same frequency, so they do not get out of alignment with one another as a result of this precessional motion. The alignment between the moments is therefore maintained.

After the 90° pulse the spins are definitely not at equilibrium. The question is, how can equilibrium be restored? One simple option is to apply another 90° pulse which, if its phase is chosen correctly, will rotate the magnetization back to its equilibrium position along the z-axis. However, this is not relaxation – this is just us manipulating the spins. Relaxation is a natural process which takes place without any intervention from us, so we must identify another way in which the magnetization can be driven to its equilibrium position.

9.1.2 Local fields

From all we have seen so far, it is clear that to rotate the magnetization towards the z-axis we need a transverse magnetic field which is oscillating at or near to the Larmor frequency. During an RF pulse we apply such a field deliberately, but it turns out that such oscillating fields also occur *naturally* within the sample. These fields can interact with the individual magnetic moments, and thus rotate them to new positions, in just the same way as an RF pulse.

There are various mechanisms which lead to the generation of magnetic fields within the sample, and we will discuss these in more detail later on. At this stage it is helpful to describe just one source of these fields, which is from other spins in the sample. Each spin has a magnetic moment which, as we have noted before, behaves like a small bar magnet, generating its own magnetic field. A spin therefore experiences not only the static applied field but also magnetic fields from nearby spins; this is illustrated in Fig. 9.3

This field generated by a spin – called the *local field* – falls off rapidly with distance, and even at the closest approach, is many orders of magnitude weaker than the applied field. The local field varies in *magnitude* and *orientation* as the molecules move around due to thermal agitation. If this motion results in the transverse component of the field oscillating at close to the Larmor frequency, a magnetic moment which experiences the local field will be rotated to a new direction, just as it would be by a pulse.

The local field acts like a pulse but, rather than all of the spins being affected in the same way, the effect is highly localized. The local field is different in different parts of the sample, so each spin is affected in a *different* way. This is in complete contrast to a pulse, which affects all of the spins in the *same* way.

As these local fields can change the orientation of individual magnetic moments, both the magnitude and orientation of the bulk magnetization will be affected. What we now need to explore is how these local fields drive the bulk magnetization to its equilibrium position.

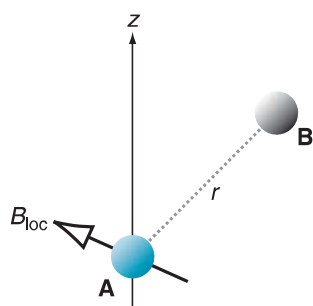


Fig. 9.3 Spin A experiences a local field, B_{loc} , due to the magnetic moment of a nearby spin B. The magnitude and direction of the local field depends on the distance r between the two spins, and the orientation of the vector joining the two spins (shown dashed) with respect to the applied field, which is along the z-axis.

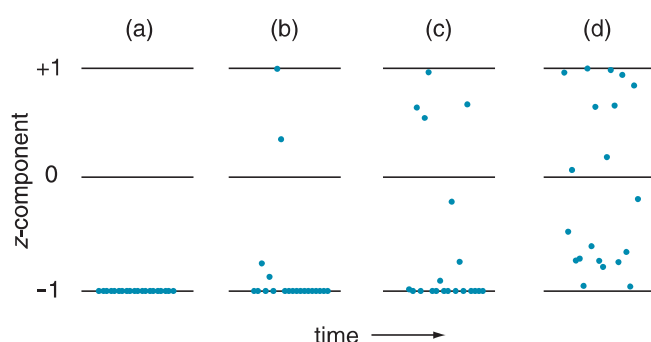


Fig. 9.4 Visualization of how the bulk z -magnetization is driven to zero by random changes in the z -component of the magnetic moments of individual spins. The z -component of each of 20 spins is represented by a dot; if the spin is aligned along $+z$, the component takes the value $+1$, whereas if the spin is aligned along $-z$, the component takes the value -1 . The initial state, shown in (a), has all of the z -components along $-z$, such that the total z -magnetization, found by adding up the individual z -components, is -20 . After a short time, some of the individual magnetic moments will be reoriented, thus changing their z -components. This is shown in (b), where the z -magnetization is -16 . If we wait longer, more of the magnetic moments will have been reoriented, giving the arrangement shown in (c), for which the z -magnetization is -12 . After sufficient time the z -magnetization will become zero, as is the case for arrangement (d). See text for discussion of why the magnetization is driven to zero, rather than to its proper equilibrium value.

If we assume, not unreasonably, that the local fields are varying randomly in their magnitude and orientation, then we expect that the bulk z -magnetization will eventually be driven to zero. The reason for this assertion is that if the individual magnetic moments are rotated by random amounts at random time intervals, after sufficient time the moments will be randomly oriented, resulting in no bulk magnetization.

This idea is illustrated in Fig. 9.4, in which we see how the z -components of just 20 spins are affected by random reorientations of the individual moments. So that we can see more clearly what is going on, at time zero the magnetic moments of all of the spins have been aligned along $-z$, as shown in (a). After some time, the magnetic moments of a few of the spins have been reoriented, leading to a change in their z -components; this is shown in (b). It is clear that the result of these reorientations is to reduce the size of the z -magnetization.

Leaving longer times, as shown in (c) and (d), results in more of the magnetic moments being reoriented, and so a further reduction in the total z -magnetization. It is clear that, after sufficient time, these random reorientations will drive the z -magnetization to zero.

There is clearly a problem with this description, as it predicts that at equilibrium there is *no* z -magnetization, which is certainly not the case. To resolve this problem with our argument we need to think about what it is that the spins are coming to equilibrium with, which is the topic of the next section.

9.1.3 Coming to equilibrium with the lattice

Recall that the local fields are varying on account of the random thermal agitation of the molecules in the sample. We describe this by saying that the local fields provide a mechanism by which the thermal motion of the molecules can be experienced by the spins – in other words it puts the spins into *contact* with the thermal motion.

We know from everyday experience that if we put two objects in contact, they will come to thermal equilibrium by exchanging heat energy. For example, if a hot lump of metal is dropped into a bucket of water the metal cools down as energy flows from the hot metal to the cooler water. As the metal cools, the water is heated, and eventually both come to the same temperature. They are then in thermal equilibrium with one another.

In our NMR sample, the nuclear spins have a certain amount of energy on account of the interaction of the magnetic moments with the applied magnetic field. Of course, the nuclei have other sorts of energy as well, but we are not concerned with these here, and so when we talk about the ‘energy of the spins’ we will just mean this energy of interaction between the spins and the field.

The interaction between the spins and the field can therefore be thought of as leading to a *reservoir* of (spin) energy. The thermal motion of the molecules is also a reservoir of energy, which is put in contact with the spin energy reservoir via the mediation of the local fields, as is illustrated in Fig. 9.5. As a result, the two reservoirs are able to come to thermal equilibrium with one another, just in the same way that two physical objects can come to thermal equilibrium.

When the bulk magnetization is rotated away from the z -axis, the energy of the spins is increased. This is because the number of spins whose magnetic moments are aligned with the z -axis, the low energy arrangement, is decreased. For the spins to come back to equilibrium they therefore need to lose energy.

The amount of energy that the spins need to lose to come to equilibrium is minuscule when compared with the energy of thermal motion. Bringing the spins back to equilibrium is analogous to dropping our hot metal into a swimming pool rather than a bucket.

The process by which the z -magnetization is returned to its equilibrium value is called *longitudinal relaxation*. You will also find it referred to as *spin–lattice relaxation*. This latter term arises from the understanding that this kind of relaxation involves the flow of energy between the spins and the molecular motion. The ‘lattice’ is a generic term used to describe a reservoir of energy, which in this case is the molecular motion.

The equilibrium z -magnetization

In the previous section we discussed a simple picture in which the reorientation of individual magnetic moments by random fields drove the bulk z -magnetization to zero – a result which is plainly wrong, as we know that relaxation results in the establishment of a finite z -magnetization at equilibrium. It is not that this picture of the magnetic moments being reoriented is wrong, it is just that there is an additional subtlety which

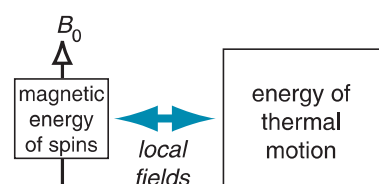


Fig. 9.5 The interaction of the spins with the applied magnetic field, and the thermal motion of the molecules, can both be thought of as leading to reservoirs of energy. The two reservoirs are able to come to equilibrium as a result of the mediation provided by the local fields i.e. the local fields create thermal contact between the two reservoirs.

we need to take account of in order to arrive at the correct equilibrium position.

If the local field at a particular spin results in its magnetic moment being rotated *towards* the z -axis, the energy of interaction between the spin and the applied field is reduced, and so there must be a flow of energy from the spin to the surroundings. In contrast, if the magnetic moment is turned *away* from the z -axis, the magnetic energy of the spin increases, and so there must be a flow of energy from the surroundings to the spin.

The key point to understand here is that there is a higher probability for a certain amount of energy to flow into the surroundings than there is for the same amount of energy to flow out of the surroundings. The reason for this is connected to the fact that the surroundings are at thermal equilibrium, a situation described by the Boltzmann distribution. This distribution tells us that as the energy of a particular state (i.e. energy level) of the surroundings goes up, the probability of that state being occupied goes down.

When energy is absorbed, the surroundings move from a lower energy state to a higher energy one; in contrast, when energy is given out, the surroundings move from a higher energy state to a lower energy one. As the lower energy state is the more occupied of the two, it follows that it is more probable that the surroundings will absorb, rather than give out, a certain amount of energy.

The overall result of this asymmetry between the probability of giving out and receiving energy is that events in which the magnetic moments are moved towards the z -axis are more probable than those in which the moment moves away from the z -axis. As a result, after many such events, the z -components of the moments are not distributed randomly, but in such a way as to lead to a net z -magnetization.

This discussion relies on the fact that the energy of interaction between the spins and the magnetic field is minuscule compared with the energy associated with the thermal motion of the molecules. We can therefore safely assume that once the surroundings are at equilibrium, the flow of energy to and from the spins will not perturb this equilibrium in any perceptible way. It also follows that the difference in the probability of these tiny amounts of energy flowing into or out of the surroundings is very small, so that the distribution of the z -components of the magnetic moments is only slightly perturbed from a random arrangement.

9.1.4 Transverse relaxation

The process by which transverse magnetization decays away to its equilibrium value of zero is called *transverse relaxation*. It is also called *spin-spin relaxation*, a confusing and unhelpful term which we will avoid.

When we talk about the individual magnetic moments changing orientation as a result of interacting with an oscillating transverse local field, it is not only the z -, but also the transverse components of the moments which can be changed. For example, if the local field has a component along x , then the y -component of the magnetic moment can be changed. Similarly, a local field along y can change the x -component. Overall therefore, these local fields not only drive the z -magnetization to its equilibrium value,

The origin of the terms *secular* and *non-secular* will be described on page 293.

but also do the same for the transverse magnetization. This is called the *non-secular* contribution to transverse relaxation.

There is, however, a second way in which the transverse magnetization is driven to equilibrium. Remember that the magnetic moments from individual spins precess about the applied magnetic field B_0 at the Larmor frequency (Fig. 9.2 on page 243). However, the field experienced by a particular spin is not just B_0 , but is this field *plus* the z -component of the local field. As a result the precession frequency is changed, albeit by rather a small amount as the local field is much smaller than B_0 .

The local field varies from spin to spin, so the precession frequency is slightly different for each spin. Consequently, over time the precession of the individual moments will get out of step with one another, and thus the net transverse magnetization will shrink. This is the *secular* contribution to transverse relaxation.

If the local fields did not change over time, then this decay of the transverse magnetization could be reversed using a spin echo. This comes about because the frequency at which each spin is evolving is constant, so a spin echo will refocus this, in just the same way that an offset term is refocused. However, it is *not* the case that the local fields are time independent – on the contrary, they are changing rapidly due to molecular motion. As a result, a spin echo cannot refocus the evolution at this ever changing frequency, and so the decay of the transverse magnetization cannot be reversed. A more detail discussion of transverse relaxation is deferred to section 9.8 on page 286.

9.1.5 Summary

Rather a lot of new ideas have been introduced in this section, so it is helpful to summarize the key steps in the discussion.

- The magnetic moment from each individual spin behaves in the same way as the bulk magnetization i.e. it can be rotated by a transverse field oscillating at the Larmor frequency, and precesses about the field along the z -axis.
- In the sample there are sources of local magnetic fields which vary randomly in orientation and magnitude. These fields are highly localized.
- If the local field experienced by a spin has a transverse component which is oscillating at the Larmor frequency, then the orientation of the magnetic moment of the spin will change. In particular, the components of the magnetic moment which are perpendicular to the transverse local field will be changed.
- These changes in orientation of the magnetic moments of individual spins result in changes in the bulk magnetization of the sample.
- The local fields provide a thermal contact between the spins and the random thermal motion of the molecules. This drives the z -magnetization back to its equilibrium value.

- The return of the z -magnetization to its equilibrium value is called longitudinal relaxation. Such relaxation is brought about by the transverse components of local magnetic fields which are oscillating at the Larmor frequency.
- The decay of transverse magnetization to its equilibrium value of zero is called transverse relaxation.
- There are two contributions to transverse relaxation: the non-secular contribution, like longitudinal relaxation, is brought about by the transverse components of local fields which are oscillating at the Larmor frequency; the secular contribution is caused by there being a distribution of the z -components of the local fields.

Our task now is to describe in more detail the various sources of these local fields, and then to go on to investigate how we can characterize the random motions which give rise to the required time dependence.

9.2 Relaxation mechanisms

A particular source of a local magnetic field is called a *relaxation mechanism*. While there are quite a lot of these, two tend to be dominant for spin-half nuclei: the *dipolar* mechanism and the *chemical shift anisotropy* mechanism. We will therefore largely confine our attention to these mechanisms.

9.2.1 The dipolar mechanism

In this mechanism the local field is due to the magnetic moment (or magnetic dipole, as it is sometimes called) of another spin, as was outlined in section 9.1.2 on page 244 and illustrated in Fig. 9.3 on the same page. There are thus two spins, or dipoles, involved: one generating the field, and one experiencing it. This is why the mechanism is also called the *dipole-dipole* mechanism.

The local field due to the neighbouring spin depends on a number of parameters:

- The distance r between the two spins: in fact, the interaction falls off rather quickly as $(1/r^3)$.
- The gyromagnetic ratio of the spin: the larger the gyromagnetic ratio, the larger the magnetic moment and the larger the local field. So, for example, protons give rise to larger local fields than ^{13}C nuclei.
- The orientation of the vector joining the two spins relative to the applied magnetic field (the z -axis).

With this strong distance dependence, the dipolar mechanism is only effective over rather short distances, say less than 5 Å for two protons. However, note that local fields can be contributed by more than one spin, so equally effective relaxation can be brought about by either a single nearby spin or a larger number of more remote spins.

The final point to note is that the effect of a given local field B_{loc} on the nucleus depends on the gyromagnetic ratio *of that nucleus*. This is because the rate at which the local field rotates the magnetic moment goes as γB_{loc} , in just the same way that an RF field B_1 causes the bulk magnetization to precess at a frequency γB_1 . Therefore overall the strength of the dipolar interaction depends on the gyromagnetic ratio of the spin which is generating the local field, and the gyromagnetic ratio of the spin which is experiencing that local field.

9.2.2 Chemical shift anisotropy

The usual description given for the chemical shift is to say that, in the presence of a strong applied field, the electrons in the molecule give rise to a small induced (local) field at the nucleus. The nucleus therefore experiences the sum of the applied field and this induced field, thus shifting the Larmor frequency by an amount which depends on the size of the induced field.

For most molecules, the size of the induced field, and hence the size of the chemical shift, depends on the orientation of the molecule with respect to the applied magnetic field. This is described by saying that the chemical shift is *anisotropic*. In liquid samples the molecules are tumbling so rapidly that the nuclei see an average local field, and hence have an average chemical shift, called the isotropic shift. Nevertheless, at any instant, the local field is different for molecules at different orientations.

What is not usually commented on is that the local field due to the chemical shift is not necessarily parallel to the applied field – in fact in general this local field can point in any direction, as illustrated in Fig. 9.6. We see that this local field is therefore a relaxation mechanism.

The extent to which the local field varies as the molecule tumbles depends on the anisotropy of the chemical shift i.e. the extent to which the shift varies with orientation. With the exception of nuclei at sites of high symmetry, such as isolated atoms or nuclei in octahedral or tetrahedral sites, all chemical shifts are anisotropic. The extent of the anisotropy does, however, vary greatly between different isotopes.

As a rough guide, the shift anisotropy is of the order of the chemical shift range for that nucleus. So, for protons the shift anisotropy is rarely more than a few ppm, whereas for ^{13}C the anisotropy can easily be 100 ppm, or more. Nuclei such as ^{31}P , which have wide chemical shift ranges, can also have large anisotropies.

The local field due to the chemical shift is proportional to the applied field. In turn the interaction of this field with the nucleus depends on the gyromagnetic ratio of the nucleus, so overall the interaction goes as γB_0 .

9.2.3 Relaxation by paramagnetic species

In the dipolar mechanism, the source of the local field is the magnetic moments of other nuclear spins in the sample. Unpaired electrons also generate local fields in an analogous way, and so are a potential source of relaxation. The magnetic moment of the electron is very much greater than that of the proton, so the local field generated by an electron is correspondingly much greater. Unpaired electrons can, therefore, be a

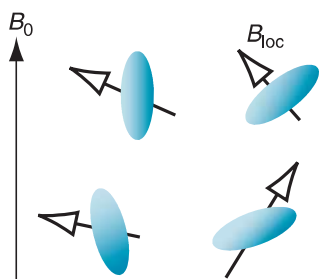


Fig. 9.6 When placed in a strong applied field B_0 , a local field is generated as a result of the interaction between the electrons and B_0 . It is this field which is responsible for the chemical shift. However, on account of the anisotropy of the electron distribution, the local field varies in direction and size as the molecule tumbles in solution. The resulting variation in the local field can be a source of relaxation.

particularly efficient source of relaxation, causing a significant effect even when present at low concentrations.

In preparing an NMR sample it is inevitable that small amounts of O_2 gas are dissolved in the solvent, and since O_2 has unpaired electron spins (i.e. it is paramagnetic), the dissolved oxygen is a source of relaxation. Sometimes, the sample is ‘degassed’, for example by bubbling pure nitrogen gas through the sample, in order to remove the oxygen and hence reduce the rate of relaxation.

9.3 Describing random motion – the correlation time

The next task is to work out how to describe and quantify the random thermal motion which provides the all-important time dependence to the local fields. Recall from our previous discussion that to cause longitudinal relaxation the transverse component of the local field must be oscillating at or near the Larmor frequency.

A molecule is likely to be executing two distinct kinds of motion: vibrations and overall rotation. During a vibration, the individual bonds and bond angles are oscillating back and forth about their equilibrium positions, and in principle this will modulate the dipolar interaction as the distance, and possibly the angle between the internuclear vector and the field direction, will change. However, such oscillations typically take place at frequencies of 10^{11} to 10^{13} Hz, which is much higher than even the highest achievable Larmor frequency of around 10^9 Hz. We can thus discount such vibrations as being effective sources of relaxation.

The overall rotation of a molecule will result in the modulation of the local fields due to both the dipolar interaction and the chemical shift anisotropy (CSA), therefore such motion can be a source of relaxation. The question is, are these motions at the right frequencies to cause relaxation?

In a gas, small molecules rotate freely at frequencies of around 10^8 to 10^9 Hz, which is certainly in the right ball-park for relaxation. However, in liquids the picture is quite different, and it is certainly not the case that even small molecules are free to rotate. The density of a liquid means that the frequency of collisions between molecules is rather high, but the high density, combined with the interactions between the molecules, means that the ability of a molecule to rotate is rather constrained. So, each collision is only capable of changing the orientation of a molecule by a small amount.

One way of visualizing this situation is to imagine that the solvent exerts a viscous drag on the molecule which is so large that even an energetic collision is only able to rotate the molecule by a small amount, if at all. An analogy would be trying to make a floating football rotate by throwing ping-pong balls at it.

Let us concentrate on just one molecule, and place an imaginary vector in it, such that the arrow starts out pointing along $+z$. As the molecule experiences collisions it will start to be rotated, in small steps, away from its starting position. However, since the collisions are random, the vector will not move steadily away from $+z$, but will execute a jerky motion in which the direction it moves, and the angle through which it moves, is different on each step. Such motion is called *rotational diffusion*.

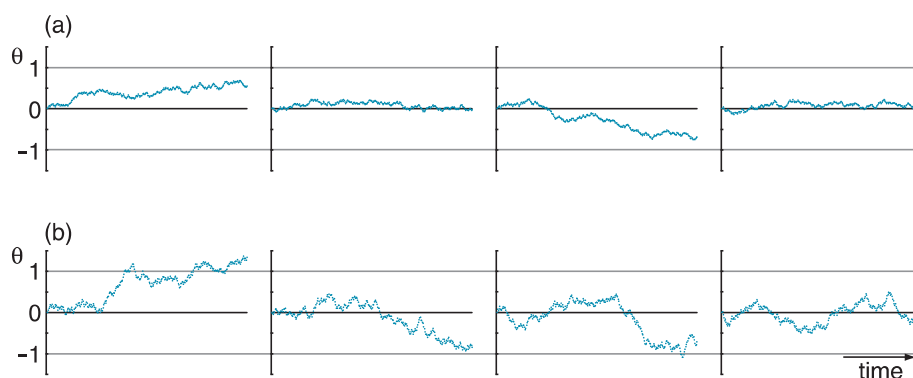


Fig. 9.7 Plots of the orientation, specified by the angle θ (rad), of a disc undergoing rotational diffusion. Each graph contains 300 equally spaced time steps, and the angle always starts at zero. In (a) the maximum allowed change in the angle on each step is ± 0.04 radians, with the actual change being chosen at random in this interval. For the plots shown in (b) the maximum change is ± 0.08 radians. Each plot (trajectory) is different on account of the random nature of the process. However, on average the trajectories shown in (b) deviate further from the starting position, as a result of the greater change in the angle allowed on each time step. The rotational correlation time for motion of which (a) is a representative sample of trajectories is therefore longer than for those shown in (b).

If we wait a ‘long time’, the vector will have wandered through more or less all possible orientations, whereas after a ‘short time’ the vector will barely have moved. What ‘long’ and ‘short’ mean can be quantified by defining a *rotational correlation time* τ_c , which is the *average* time it takes for a molecule to end up at an orientation about 1 radian from its starting position. Remember that the molecule does not jump to this new position in a few steps, but takes a tortuous and wandering path of many tiny steps before it finally finds itself 1 radian from where it started.

Figure 9.7 provides a visualization of the nature of rotational diffusion and the meaning of the correlation time. To make this diagram, we have simplified things by considering a two-dimensional case, for which only one angle is needed to specify the orientation i.e. we are thinking about the rotational diffusion of a disc rather than a sphere. Each graph shows the angle for 300 time steps, where at each step the angle is allowed to change by a random amount.

For the plots shown in (a), the maximum change in the angle allowed on any one step is ± 0.04 radians, whereas in (b) the maximum change is ± 0.08 radians. On any one time step, the size of the change is chosen at random within the permitted range.

The first thing to note is that, although the maximum change in the angle allowed on each step was the same for all the plots shown in (a), the actual sequence of angles is different in each case. This is expected as it is a random process, so each molecule heads off on its own unique trajectory. In none of the trajectories does θ reach one radian, so the correlation time must be somewhat longer than the time represented by these 300 steps.

For the plots shown in (b), the maximum change in the angle on each step is twice that for (a), so not surprisingly the excursions away from the

starting position are greater. In fact, within the time frame the angle reaches 1 radian for two of the trajectories, and almost makes it for a third. It is clear that the correlation time for this set of trajectories must be shorter than that for those shown in (a).

It is important to realize that the correlation time is the *average* time needed to achieve an orientation 1 radian away from the starting position, not the time taken by one particular molecule to achieve this orientation. This average is taken over a large number of molecules in the sample.

For small molecules τ_c turns out to be around 10 ps, rising to 10 ns for small proteins. The reciprocal of the correlation time, $1/\tau_c$, is a rough estimate of the ‘average frequency’ of the motion (in rad s^{-1}), so $1/(2\pi\tau_c)$ gives this frequency in Hz. Correlation times in the range 10 ns to 10 ps give average frequencies in the range of 10^7 to 10^{10} Hz, which is comparable with typical Larmor frequencies. Rotational diffusion therefore appears to give motion in the right range to cause relaxation.

The problem we have is in trying to quantify the amount of this motion which is actually at the right frequency – that is, at the Larmor frequency. To do this we need to introduce the *correlation function* and the *spectral density*.

9.3.1 The correlation function

The correlation function is a way of characterizing the time dependence of the random motion in our sample, and hence finding out how much of the motion is present at the Larmor frequency. We will start out by describing what the correlation function is, and then, in the following section, explain how this function can be used to assess the amount of motion at particular frequencies.

Imagine that a particular spin i experiences a local field $B_{\text{loc},i}(t)$ which is varying in time, so that at a time τ later the local field is $B_{\text{loc},i}(t + \tau)$. The correlation function $G(t, \tau)$ is defined as the average over the sample of the product $B_{\text{loc}}(t)B_{\text{loc}}(t + \tau)$:

$$\begin{aligned} G(t, \tau) &= \frac{1}{N} [B_{\text{loc},1}(t)B_{\text{loc},1}(t + \tau) + B_{\text{loc},2}(t)B_{\text{loc},2}(t + \tau) + \dots] \quad (9.1) \\ &= \frac{1}{N} \sum_{i=1}^N B_{\text{loc},i}(t)B_{\text{loc},i}(t + \tau) \\ &= \overline{B_{\text{loc}}(t)B_{\text{loc}}(t + \tau)}, \end{aligned}$$

where N is the number of spins in the sample. On the last line we have used the overbar to indicate the average over the sample, known as the ensemble average.

Generally $B_{\text{loc}}(t)$ can take positive or negative values, which are equally distributed either side of zero, so that its average over the sample is zero. It is also usually the case that the magnitude of the local field from a particular source has a maximum value.

The local field varies due to the thermal motion in the sample, so $B_{\text{loc}}(t)$ is a random function of time. It turns out that the properties of this random function do not depend on the point from which time is measured; such functions are said to be *stationary* random functions. For such a random

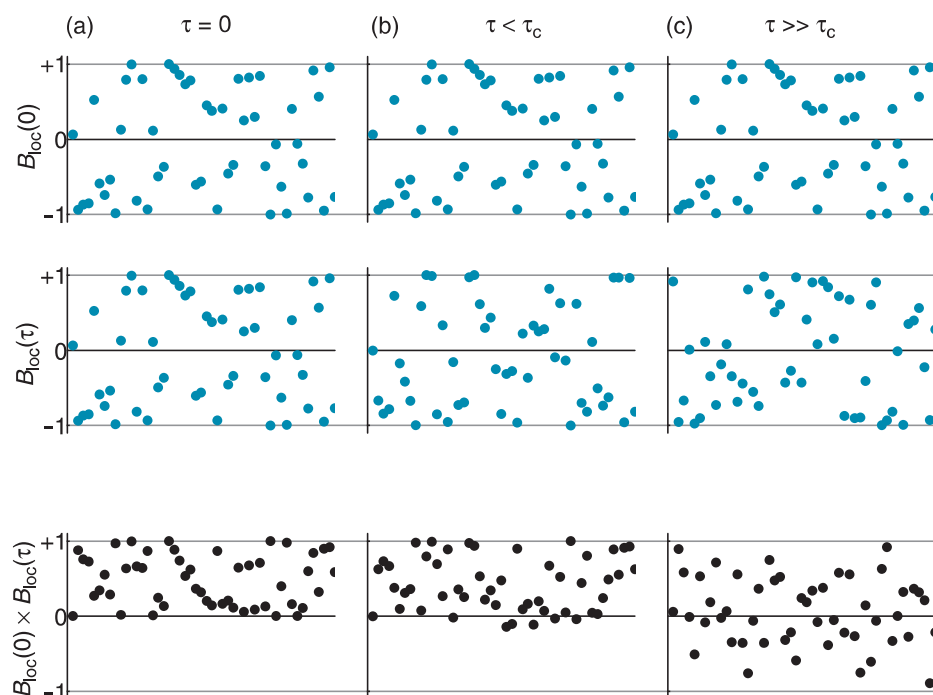


Fig. 9.8 Visualization of the behaviour of the correlation function $G(\tau)$ as a function of the time interval τ . In each of (a), (b) and (c), the data points represent: in the top row, $B_{\text{loc}}(0)$; in the middle row, $B_{\text{loc}}(\tau)$; and in the bottom row, the product $B_{\text{loc}}(0)B_{\text{loc}}(\tau)$. The values for each of 50 spins are represented by a dot, and it has arbitrarily been assumed that B_{loc} can only take values between -1 and $+1$. Note that the horizontal axis is *not* time, but is just used to spread out the values from the 50 spins. $G(\tau)$ is found by summing the points $B_{\text{loc}}(0)B_{\text{loc}}(\tau)$ (shown in the bottom row), and dividing by the number of points. In (a) $\tau = 0$, so $B_{\text{loc}}(\tau) = B_{\text{loc}}(0)$, and as a consequence all of the points in the lower plot are positive. $G(\tau)$ is therefore a maximum when $\tau = 0$. In (b) τ is shorter than correlation time τ_c , and so although for each spin $B_{\text{loc}}(\tau)$ is different to $B_{\text{loc}}(0)$, the change is not large. Most of the points in the lower plot are positive, but in the few cases where the local field has changed sign between time 0 and time τ , the point is negative, and as a result $G(\tau)$ is less than $G(0)$ (in this case $0.87 \times G(0)$). Finally, in (c) $\tau \gg \tau_c$, so that the local field of each spin has changed significantly between time zero and time τ . As a result, there are a significant number of negative points in the lower plot, and so $G(\tau)$ is significantly less than $G(0)$.

function, the value of the correlation function does not therefore depend on the time t , but only on the time *interval* τ . So, from now on we will write the correlation function as $G(\tau)$.

Figure 9.8 helps us to visualize how the correlation function varies with τ . The points in this figure represent the values of $B_{\text{loc}}(0)$, $B_{\text{loc}}(\tau)$, and the product $B_{\text{loc}}(0)B_{\text{loc}}(\tau)$ for different values of τ , and for 50 spins. For each value of τ , the top graph shows $B_{\text{loc}}(0)$, the middle shows $B_{\text{loc}}(\tau)$, and the lower shows $B_{\text{loc}}(0)B_{\text{loc}}(\tau)$. It follows from the definition of $G(\tau)$, Eq. 9.1 on the preceding page, that the value of $G(\tau)$ is found by summing the values of $B_{\text{loc}}(0)B_{\text{loc}}(\tau)$ in the lower plot and then dividing by the number of values.

The plots shown under (a) are for the case $\tau = 0$. Here $B_{\text{loc}}(0)B_{\text{loc}}(\tau) = B_{\text{loc}}^2(0)$, so all of the points in the lower plot are positive, and hence $G(\tau)$ will be a maximum. In general, we can see that it will always be the case that the maximum value of $G(\tau)$ occurs at $\tau = 0$.

The plots shown under (b) are computed for a value of τ which is shorter than the correlation time. Careful comparison of the plots of $B_{\text{loc}}(0)$ and $B_{\text{loc}}(\tau)$ will show that, although the local fields have changed between the two times, the change is small. As a result $B_{\text{loc}}(0)B_{\text{loc}}(\tau)$ is still positive for most spins, but there are a few where the local field has changed sign, and for these $B_{\text{loc}}(0)B_{\text{loc}}(\tau)$ is negative. So, when $G(\tau)$ is found by computing the sum of the points in the lower plot the result will be somewhat less than $G(0)$.

Finally, the plots shown under (c) are for the case where the time τ is much longer than the correlation time. Now the local fields have changed significantly between time zero and time τ , so there are a significant number of negative points in the lower plot. Summing these points, that is computing $G(\tau)$, gives a value which is much less than $G(0)$.

Our overall picture for how the correlation function varies with τ is that it starts at a maximum at $\tau = 0$, and then falls away smoothly to zero at a rate determined by the value of the correlation time τ_c . The value of $G(0)$ can be computed from the definition of $G(\tau)$ given in Eq. 9.1 on page 253:

$$\begin{aligned} G(0) &= \overline{B_{\text{loc}}(t)B_{\text{loc}}(t)} \\ &= \overline{B_{\text{loc}}^2}. \end{aligned}$$

This value is simply the average of the square of the local field. The time at which we compute this average is irrelevant for a stationary random function, so $G(0)$ just depends on the average size of the interaction.

The exact form of the correlation function depends on the details of the interaction between the molecule and the solvent. The simplest case to analyse is where the molecule is assumed to be spherical and the solvent simply provides a medium with a certain viscosity. Rotational diffusion in such a case, turns out to be described by a correlation function which is a simple exponential whose decay rate is set by τ_c :

$$G(\tau) = \overline{B_{\text{loc}}^2} \exp(-|\tau|/\tau_c).$$

A plot of this function for three different correlation times is shown in Fig. 9.9. Note that the smaller the correlation time, the faster $G(\tau)$ decays. The function depends on the modulus of the time τ , as the same behaviour is expected for positive times τ as it is for negative times i.e. the behaviour for a particular time interval in the future is the same as for the same interval of time in the past.

The exponential part of the correlation function is independent of the source of the local fields, which only determines the overall magnitude of $G(\tau)$ via the term $\overline{B_{\text{loc}}^2}$. It is therefore common to define a *reduced correlation function*, $g(\tau)$, which is independent of the size of the local fields:

$$g(\tau) = \exp(-|\tau|/\tau_c). \quad (9.2)$$

With this definition, $G(\tau) = \overline{B_{\text{loc}}^2} g(\tau)$.

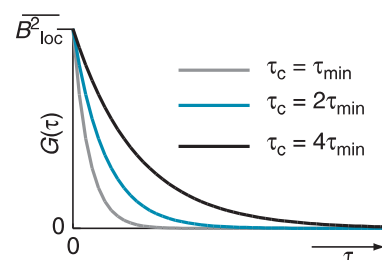


Fig. 9.9 Plot of the correlation function $G(\tau) = \overline{B_{\text{loc}}^2} \exp(-|\tau|/\tau_c)$ for three different values of the correlation time, τ_c . The curve for shortest value of τ_c , τ_{min} , is shown in grey, the blue line is for a τ_c of twice this value, and the black line is for twice this value again. Note that for all three cases, $G(\tau)$ has the same maximum value of $\overline{B_{\text{loc}}^2}$ at $\tau = 0$.

9.3.2 The spectral density

From the discussion in section 9.1.2 on page 244, you will recall that, in order to be effective at causing longitudinal relaxation, the local field must be oscillating close to the Larmor frequency. We have now seen that the time dependence of the local field results from molecular motion, specifically rotational diffusion, and that this motion can be characterized by a correlation function. The next task is to find out exactly how much of the motion is present at the required frequency. We will see in this section that the *spectral density* function, which is related to the correlation function, provides this information.

The correlation function is a function of *time*, analogous to the FID we observe in an NMR experiment. If we Fourier transform the FID we obtain a function of frequency, the spectrum, which tells us how much intensity there is at each frequency. In an analogous way, **if we Fourier transform the correlation function we will obtain a function of frequency**, and from this we will be able to find the amount of motion which is at the Larmor frequency.

The Fourier transform of the correlation function is called the *spectral density*, $J(\omega)$:

$$G(\tau) \xrightarrow{\text{FT}} J(\omega).$$

The amount of motion at the Larmor frequency is simply found by evaluating $J(\omega)$ at $\omega = \omega_0$.

In the case of the exponential correlation function, the Fourier transform is the familiar absorption mode Lorentzian, centred at $\omega = 0$:¹

$$\overline{B_{\text{loc}}^2} \exp(-|\tau|/\tau_c) \xrightarrow{\text{FT}} \overline{B_{\text{loc}}^2} \frac{2\tau_c}{1 + \omega^2\tau_c^2},$$

so

$$J(\omega) = \overline{B_{\text{loc}}^2} \frac{2\tau_c}{1 + \omega^2\tau_c^2}. \quad (9.3)$$

Figure 9.10 shows plots of this function for different values of the correlation time; only positive frequencies are shown. The spectral density has its maximum value of $2\overline{B_{\text{loc}}^2}\tau_c$ at $\omega = 0$, and then falls off steadily as ω increases, with the rate of the fall-off being determined by τ_c . As τ_c becomes shorter, the spectral density spreads out to higher frequencies. Note also that the value of $J(0)$ increases as τ_c increases.

A particularly important feature of the **spectral density is that, if we plot it against ω , we find that the area under the curve is independent of τ_c** . Expressed mathematically, the area under $J(\omega)$ is the integral

$$\begin{aligned} \text{area} &= \int_0^\infty \overline{B_{\text{loc}}^2} \frac{2\tau_c}{1 + \omega^2\tau_c^2} d\omega \\ &= \pi \overline{B_{\text{loc}}^2}. \end{aligned}$$

A consequence of the area being constant is that, although $J(\omega)$ is always a maximum at $\omega = 0$, as τ_c becomes smaller the size of the maximum

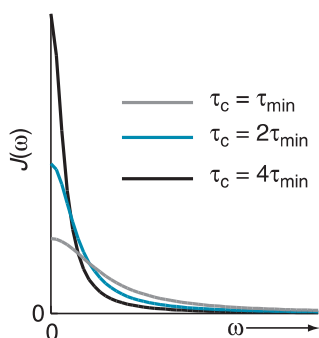


Fig. 9.10 Plots of the spectral density function, $J(\omega)$, for three different values of the correlation time, τ_c ; the graph is plotted only for positive values of ω . The grey line is for the shortest value of τ_c , τ_{min} , the blue line for the case where $\tau_c = 2\tau_{\text{min}}$, and the black line for the case where $\tau_c = 4\tau_{\text{min}}$. Note that the shorter τ_c becomes, the higher the frequencies to which $J(\omega)$ spreads. As explained in the text, it turns out that the areas under these curves are all the same.

¹The factor of two comes in as an FID only exists for time greater than zero, whereas the correlation function extends symmetrically either side of $\tau = 0$.

decreases and at the same time the spectral density spreads out to higher frequencies. This can be seen in Fig. 9.10 on the preceding page.

This behaviour has an important consequence if we think about the value of the spectral density at the Larmor frequency as a function of the correlation time. Remember that it is $J(\omega_0)$ which will determine the rate of longitudinal relaxation.

The spectral density at the Larmor frequency is given by

$$J(\omega_0) = \overline{B_{\text{loc}}^2} \frac{2\tau_c}{1 + \omega_0^2 \tau_c^2}.$$

If we make a plot of this as a function of τ_c , as is shown in Fig. 9.11, we see that there is clearly a value of the correlation time which makes $J(\omega_0)$ a maximum. It is a relatively easy calculation to show that this maximum occurs when $\omega_0 \tau_c = 1$ i.e. $\tau_c = 1/\omega_0$.

It follows that the rate of longitudinal relaxation will also be a maximum when $\tau_c = 1/\omega_0$, as this is the value of the correlation time which gives the maximum spectral density at the Larmor frequency. So, both correlation times which are shorter or longer than this optimum value will give slower relaxation. Just like Goldilocks and the porridge, for the most efficient longitudinal relaxation the correlation time must be neither too long nor too short, but ‘just right’.

In the previous section we introduced (Eq. 9.2 on page 255) the idea of the reduced correlation function $g(\tau)$, which does not depend on the size of the local fields. The Fourier transform of this reduced correlation function is the *reduced spectral density*, $j(\omega)$:

$$\begin{aligned} g(\tau) &\xrightarrow{\text{FT}} j(\omega) \\ \exp(-|\tau|/\tau_c) &\xrightarrow{\text{FT}} \frac{2\tau_c}{1 + \omega^2 \tau_c^2}. \end{aligned} \quad (9.4)$$

With this definition, it follows that $J(\omega) = \overline{B_{\text{loc}}^2} j(\omega)$.

9.3.3 Motional regimes

The comparison between the Larmor frequency and the correlation time turns out to be rather important in the theory of relaxation – indeed, we have already seen an example of this in Fig. 9.11 where the maximum in $J(\omega_0)$ occurs when $\tau_c = 1/\omega_0$.

In making this comparison it is usual to distinguish two *motional regimes*. The first is called *fast motion* or *extreme narrowing*, and is defined mathematically as when $\omega_0 \tau_c \ll 1$. Physically, this is the limit in which motion is very fast i.e. the correlation time is very short, such as would be the case for small molecules.

The reduced spectral density at the Larmor frequency is given by

$$j(\omega_0) = \frac{2\tau_c}{1 + \omega_0^2 \tau_c^2}.$$

If the fast motion limit applies, $\omega_0 \tau_c \ll 1$, it follows that $(1 + \omega_0^2 \tau_c^2) \approx 1$, and so $j(\omega_0)$ is given by

$$\text{fast motion: } j(\omega_0) = 2\tau_c.$$

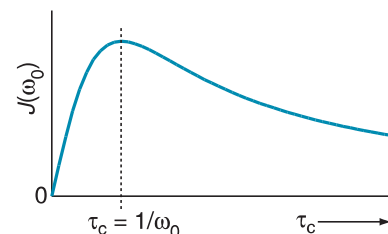


Fig. 9.11 A plot of the spectral density at the Larmor frequency, $J(\omega_0)$, as a function of the correlation time, τ_c . The plot shows a maximum at $\tau_c = 1/\omega_0$. This implies that the rate of longitudinal relaxation will be a maximum for this value of the correlation time.

In words, what this fast motion limit means is that the spectral density is independent of the Larmor frequency. It should also be noted that as $j(0) = 2\tau_c$ (for all values of τ_c), in the fast motion limit it follows that $j(\omega_0) = j(0)$.

The other limit is called the *slow motion* or *spin diffusion* limit, and is when $\omega_0\tau_c \gg 1$. In this limit, $(1 + \omega_0^2\tau_c^2) \approx \omega_0^2\tau_c^2$, so the reduced spectral density becomes

$$\text{slow motion: } j(\omega_0) = \frac{2}{\omega_0^2\tau_c}.$$

Recalling that $j(0) = 2\tau_c$, this expression can be written

$$j(\omega_0) = \frac{j(0)}{\omega_0^2\tau_c^2}.$$

As $\omega_0^2\tau_c^2 \gg 1$, it follows that $j(\omega_0)$ is very much smaller than $j(0)$.

For a Larmor frequency of 500 MHz ($\omega_0 = 3.1 \times 10^9 \text{ rad s}^{-1}$), a small molecule with a correlation time of 10 ps has $\omega_0\tau_c = 0.03$, which is clearly in the fast motion limit. In contrast, a small protein with a correlation time of 10 ns has $\omega_0\tau_c = 31$, which is clearly in the slow motion limit.

9.3.4 Summary

To summarize what we have found so far:

- Rotational diffusion in a liquid causes the local fields to vary at rates which are suitable for causing relaxation.
- This random diffusive motion can be described using a correlation time τ_c , which is the average time it takes for a molecule to move to a position at an angle of about 1 radian from its starting position.
- The spectral density gives the frequency distribution of the motion. The simplest example of such a spectral density function is

$$J(\omega) = \overline{B_{\text{loc}}^2} \frac{2\tau_c}{1 + \omega^2\tau_c^2}.$$

Note that $J(\omega)$ depends on the correlation time, τ_c .

- The rate of longitudinal relaxation depends on the spectral density at the Larmor frequency. The relaxation is most rapid when $\omega_0\tau_c = 1$.

We are now in a position to explore the details of how the z -magnetization behaves as a result of relaxation. To do this, we will first need to introduce the concept of the populations of energy levels.

9.4 Populations

In section 4.1 on page 47 we were at pains to point out that the magnetic moment from a single spin-half can point in any direction, and that its energy depends on the angle between the magnetic moment and the applied

field. However, if we measure the energy of a particular spin, we will always find a value which corresponds to one of the two energy levels of a spin-half i.e. that corresponding to the α state or the β state (see section 3.1 on page 24 for a discussion of this point). It turns out that there is a certain probability p_α of finding the energy corresponding to the α state, and a certain probability p_β of finding the energy corresponding to the β state.

If we imagine measuring the energy of every spin in the sample, then for each there is a probability $p_{\alpha,i}$ of finding the energy corresponding to the state α . The sum of all these probabilities is the same as the number of spins in the sample which were found to have the energy of the α state. This number can be identified as the *population* of this state, n_α :

$$n_\alpha = p_{\alpha,1} + p_{\alpha,2} + p_{\alpha,3} \dots$$

In a similar way, we can compute the population of the β state, n_β .

This language is rather dangerous, as the moment we start talking about the ‘populations of the α and β states’, it is easy to fall into the trap of imagining that each spin is in one of these states, which is certainly not true. Nevertheless, thinking about the sum of these individual probabilities as a population is a very useful concept when it comes to describing relaxation and so we will use it throughout the rest of this chapter.

9.4.1 The z -magnetization in terms of populations

If we accept this description of the spins in terms of the populations of the α and β states, it is easy to determine the bulk z -magnetization. A spin in the α state contributes $+\frac{1}{2}\hbar\gamma$ to magnetization, and one in the β state contributes $-\frac{1}{2}\hbar\gamma$, so the bulk z -magnetization, M_z , is

$$M_z = \frac{1}{2}\hbar\gamma(n_\alpha - n_\beta), \quad (9.5)$$

where n_α and n_β are the populations of the α and β states, respectively. What this equation says is that the magnetization is proportional to the population difference and to the gyromagnetic ratio. This latter factor comes about because the magnetic moment of each individual spin is proportional to its gyromagnetic ratio.

At equilibrium we know that the populations must be those given by the Boltzmann distribution, which in this case predicts the following:²

$$n_\alpha^0 = \frac{1}{2}N \exp(-E_\alpha/k_B T) \quad n_\beta^0 = \frac{1}{2}N \exp(-E_\beta/k_B T), \quad (9.6)$$

where E_α and E_β are the energies of the α and β states, k_B is Boltzmann’s constant, and n_α^0 and n_β^0 are the equilibrium populations.

The energies E_α and E_β are tiny compared with $k_B T$, so the fraction $(E_{\alpha/\beta}/k_B T)$ is very much less than one. We can therefore approximate the exponential by taking just the first two terms of the series expansion

$$\exp(-x) \approx 1 - x \quad x \ll 1.$$

²The factor of $\frac{1}{2}$ in these expressions is $1/q$, where q is the partition function. In this case $q = 2$, as there are two accessible states.

Using this, the populations are

$$n_{\alpha}^0 = \frac{1}{2}N(1 - E_{\alpha}/k_{\text{B}}T) \quad n_{\beta}^0 = \frac{1}{2}N(1 - E_{\beta}/k_{\text{B}}T),$$

and hence the equilibrium z -magnetization, M_z^0 , is

$$\begin{aligned} M_z^0 &= \frac{1}{2}\hbar\gamma(n_{\alpha}^0 - n_{\beta}^0) \\ &= \frac{1}{4}\hbar\gamma N \frac{E_{\beta} - E_{\alpha}}{k_{\text{B}}T}. \end{aligned}$$

Not surprisingly, the population difference depends on the energy difference between the two levels. The energies of these two levels were given in section 3.2.7 on page 30 as

$$E_{\alpha} = -\frac{1}{2}\hbar\gamma B_0 \quad E_{\beta} = +\frac{1}{2}\hbar\gamma B_0,$$

where we have included a factor of $\hbar$ so that the energies are in Joules, rather than in rad s^{-1} . Using these, the equilibrium magnetization can be written

$$M_z^0 = \frac{\gamma^2 \hbar^2 N B_0}{4k_{\text{B}}T}. \quad (9.7)$$

This expression says that the bulk magnetization is proportional to the number of spins, the square of the gyromagnetic ratio, and the strength of the applied field. Consequently the largest equilibrium magnetization, and hence the strongest signals, come from nuclei with the greatest gyromagnetic ratios and from using the highest magnetic field strengths.

The expression for M_z^0 given in Eq. 9.7 goes as γ^2 . This is because the energies of the states, which determine their populations, are proportional to γ , and the magnetic moment contributed by each spin is also proportional to γ .

Since it is not possible to measure that absolute size of an NMR signal, the absolute size of the z -magnetization is not important in our calculations. As a result it is usually acceptable to drop all the constants in Eq. 9.5 on the previous page and simply write

$$M_z = n_{\alpha} - n_{\beta} \quad \text{and} \quad M_z^0 = n_{\alpha}^0 - n_{\beta}^0.$$

Although these expressions are not correct in a formal sense, they capture the essence of the situation, which is that the magnetization is proportional to the population difference.

9.4.2 Relaxation in terms of populations

Longitudinal relaxation drives the z -magnetization towards its equilibrium value, and we have now seen that the z -magnetization is proportional to the population difference between the α and β energy levels. It therefore follows that the approach to equilibrium involves changing the populations of the two levels, so that longitudinal relaxation can be described as arising from *transitions* between the two levels. For example, a transition from α to β will decrease the population difference and hence reduce the z -magnetization, whereas a transition from β to α will increase the population difference, and hence increase the z -magnetization.

The simplest assumption we can make is that the rate of transitions from α to β is proportional to the population of the α state, n_α :

$$\text{rate from } \alpha \text{ to } \beta = W_{\alpha \rightarrow \beta} n_\alpha,$$

where we have written the constant of proportion as $W_{\alpha \rightarrow \beta}$. This constant is in fact a rate constant, entirely analogous to a first-order rate constant in kinetics; both have units of (time)⁻¹ e.g. s⁻¹.

Similarly, the rate of transitions from β to α is

$$\text{rate from } \beta \text{ to } \alpha = W_{\beta \rightarrow \alpha} n_\beta.$$

Note that this time the rate is proportional to the population of the β state, as it is from this state that the transitions originate, and that we have also allowed the rate constant to be different. The two processes are illustrated in Fig. 9.12.

Transitions from α to β *decrease* the population of the α state, whereas transitions from β to α *increase* the population of the α state. Therefore the overall rate of change of the population of the α state is:

$$\text{rate of change of } n_\alpha = \underbrace{+W_{\beta \rightarrow \alpha} n_\beta}_{\text{increase in } n_\alpha} - \underbrace{W_{\alpha \rightarrow \beta} n_\alpha}_{\text{decrease in } n_\alpha}. \quad (9.8)$$

The first term is positive as it describes the rate of a process which increases the population of the α state, whereas the second term is negative as it describes a process which reduces the population of the state.

We can write an analogous equation for the rate of change of the population of the β state:

$$\text{rate of change of } n_\beta = \underbrace{+W_{\alpha \rightarrow \beta} n_\alpha}_{\text{increase in } n_\beta} - \underbrace{W_{\beta \rightarrow \alpha} n_\beta}_{\text{decrease in } n_\beta}. \quad (9.9)$$

At equilibrium the populations will be constant i.e. the rate of change will be zero. In addition, these populations will have their equilibrium values, as predicted by Eq. 9.6 on page 259. Applying these two conditions to the expressions of Eqs 9.8 and 9.9, we have

$$0 = W_{\beta \rightarrow \alpha} n_\beta^0 - W_{\alpha \rightarrow \beta} n_\alpha^0 \quad 0 = +W_{\alpha \rightarrow \beta} n_\alpha^0 - W_{\beta \rightarrow \alpha} n_\beta^0.$$

From either of these it follows that

$$\frac{n_\alpha^0}{n_\beta^0} = \frac{W_{\beta \rightarrow \alpha}}{W_{\alpha \rightarrow \beta}}. \quad (9.10)$$

There is a difficulty with this expression, in that the theory most commonly used to predict values for the rate constants $W_{\beta \rightarrow \alpha}$ and $W_{\alpha \rightarrow \beta}$ comes up with the answer that these two are *equal*. As a result, it follows from Eq. 9.10 that $n_\alpha^0 = n_\beta^0$ i.e. the equilibrium populations are equal, which is certainly not correct.

The problem here turns out to be a deficiency in the theory, in which the spins are treated quantum mechanically, but the surroundings (the lattice) are treated classically. To obtain the correct result, consistent with the

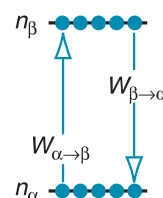


Fig. 9.12 The population of the energy level associated with the spin being in the α state, n_α , will be decreased by transitions from the α to the β state, and increased by transitions in the reverse sense. The (first-order) rate constants for these two processes are $W_{\alpha \rightarrow \beta}$ and $W_{\beta \rightarrow \alpha}$, respectively.

expected equilibrium populations, a much more complete theory is needed which treats both the spins and the lattice quantum mechanically. However, such an approach is both complicated and laborious.

What we are going to do is avoid this problem entirely by rewriting the rate equations, Eq. 9.8 and Eq. 9.9 on the previous page, in the following way:

$$\begin{aligned}\text{rate of change of } n_\alpha &= W_{\alpha\beta} (n_\beta - n_\beta^0) - W_{\alpha\beta} (n_\alpha - n_\alpha^0) \\ \text{rate of change of } n_\beta &= -W_{\alpha\beta} (n_\beta - n_\beta^0) + W_{\alpha\beta} (n_\alpha - n_\alpha^0).\end{aligned}\tag{9.11}$$

What we have done is made all the rate constants equal ($W_{\alpha\beta}$), and instead of these multiplying the populations they multiply the *deviation* of the populations from their equilibrium values i.e. $(n_\alpha - n_\alpha^0)$ and $(n_\beta - n_\beta^0)$. You can see immediately from Eq. 9.11 that, if $n_\alpha = n_\alpha^0$ and $n_\beta = n_\beta^0$, the rate of change of both n_α and n_β are zero, as is required at equilibrium.

As the z -magnetization is equal to the population difference $(n_\alpha - n_\beta)$, the rate of change of the z -magnetization is equal to the difference between the rate of change of n_α and that of n_β :

$$\text{rate of change of } M_z = \text{rate of change of } n_\alpha - \text{rate of change of } n_\beta$$

We can then use Eq. 9.11 to substitute in expressions for the rates of change of the populations:

$$\begin{aligned}\text{rate of change of } M_z &= 2W_{\alpha\beta}(n_\beta - n_\beta^0) - 2W_{\alpha\beta}(n_\alpha - n_\alpha^0) \\ &= -2W_{\alpha\beta}[(n_\alpha - n_\beta) - (n_\alpha^0 - n_\beta^0)] \\ &= -2W_{\alpha\beta}(M_z - M_z^0).\end{aligned}$$

This equation predicts what we expect: when $M_z = M_z^0$ the rate of change is zero.

The usual way to write this equation is using the language of calculus, in which we identify the rate of change of M_z as its derivative with respect to time, dM_z/dt :

$$\frac{dM_z(t)}{dt} = -R_z [M_z(t) - M_z^0].\tag{9.12}$$

We have written the z -magnetization as $M_z(t)$ to remind ourselves that it is a function of time. R_z , the rate constant for longitudinal relaxation, is equal to $2W_{\alpha\beta}$.

This equation is often written

$$\frac{dM_z(t)}{dt} = -\frac{1}{T_1} [M_z(t) - M_z^0],$$

where T_1 is the *time constant* for longitudinal relaxation: $T_1 = 1/R_z$. The use of the symbol T_1 for this time constant is so ubiquitous that longitudinal relaxation is often called ' T_1 relaxation'.

The rate of relaxation, and hence the value of the relaxation rate constant R_z , will depend, as we have seen, on the average of the square of the local fields, and the spectral density at the Larmor frequency. For the moment we will not look in detail at the calculations which lead to a prediction of the rate constant, but rather look at the implications of Eq. 9.12 for the behaviour of the z -magnetization.

9.5 Longitudinal relaxation behaviour of isolated spins

In this section we are going to investigate the practical consequences of the fact that the z -magnetization relaxes according to

$$\frac{dM_z(t)}{dt} = -R_z [M_z(t) - M_z^0].$$

This differential equation says that the rate of change of the z -magnetization is proportional to the deviation of the magnetization from its equilibrium value. The minus sign ensures that the z -magnetization changes in such a way that it approaches its equilibrium value as time increases. The rate constant R_z determines the rate at which equilibrium is approached. All this can best be seen by using an example.

Suppose that at time zero the z -magnetization is $M_z(0)$. What we will now work out, by integrating the above differential equation, is precisely how the z -magnetization moves back to its equilibrium position. We start by separating out the parts of the differential equation so that the variable $M_z(t)$ is on the left, and time is on the right:

$$\begin{aligned} \frac{dM_z(t)}{dt} &= -R_z [M_z(t) - M_z^0] \\ \text{hence } \frac{1}{[M_z(t) - M_z^0]} dM_z(t) &= -R_z dt. \end{aligned}$$

If you are not familiar with the method used to solve this kind of differential equation, then just skip forward to the result, Eq. 9.15 on the next page.

We can now integrate the left-hand side with respect to $M_z(t)$ and the right-hand side with respect to t :

$$\begin{aligned} \int \frac{1}{[M_z(t) - M_z^0]} dM_z(t) &= \int -R_z dt \\ \ln(M_z(t) - M_z^0) &= -R_z t + \text{const.} \end{aligned} \quad (9.13)$$

To compute the integrals we have used the fact that M_z^0 and R_z are constants. As usual, a constant of integration has been included, but we can find out its value by noting that at $t = 0$ the magnetization is $M_z(0)$. So at $t = 0$ Eq. 9.13 becomes

$$\ln(M_z(0) - M_z^0) = \text{const.}$$

Putting this value of the constant back into Eq. 9.13 we have

$$\ln(M_z(t) - M_z^0) = -R_z t + \ln(M_z(0) - M_z^0).$$

Tidying this up gives

$$\ln\left(\frac{M_z(t) - M_z^0}{M_z(0) - M_z^0}\right) = -R_z t. \quad (9.14)$$

If we take exponentials of both sides we have

$$\frac{M_z(t) - M_z^0}{M_z(0) - M_z^0} = \exp(-R_z t),$$

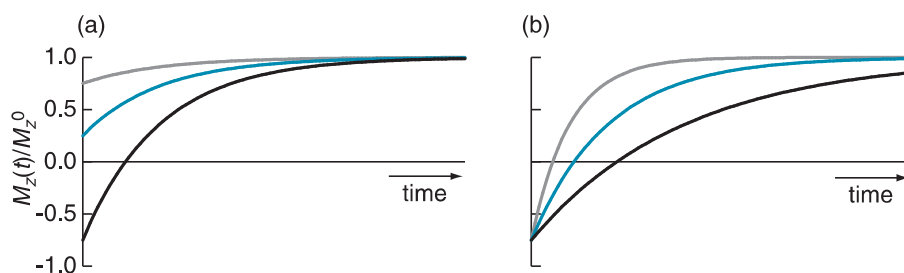


Fig. 9.13 Plots showing the way in which the z -magnetization approaches equilibrium, starting from a particular value, as predicted by Eq. 9.15. The quantity plotted along the vertical axis is $M_z(t)/M_z^0$, which takes the value one when the z -magnetization is at equilibrium. In (a), the three lines are computed for different values of the z -magnetization at time zero. Note that the further the system is from equilibrium, the more rapidly the magnetization changes. The plot in (b) shows the effect of increasing the rate constant R_z . The black line has the smallest value of R_z , the blue line has twice the value of the rate constant, and the grey line twice the value again. As expected, the greater the rate constant, the more rapidly the z -magnetization approaches its equilibrium value.

which can be rearranged to

$$M_z(t) = [M_z(0) - M_z^0] \exp(-R_z t) + M_z^0. \quad (9.15)$$

Figure 9.13 shows plots of $M_z(t)$, as predicted by this equation, for different initial z -magnetizations, $M_z(0)$, and for different values of the rate constant R_z . The common feature of all of these plots is that eventually the magnetization ends up at its equilibrium value. However, as shown in Fig. 9.13 (a), the further the magnetization is away from equilibrium, the faster the initial rate of change of the magnetization. Also, as shown in (b), the time that it takes the magnetization to reach its equilibrium value becomes shorter as the rate constant R_z becomes larger. This behaviour is in agreement with the qualitative predictions we made earlier.

9.5.1 Estimating the rate constant for longitudinal relaxation

It is often important to have an estimate for the rate constant for longitudinal relaxation, R_z , and we will see in this section that the analysis which led to Eq. 9.15 provides a convenient framework for doing this.

The method most commonly used for estimating R_z is the *inversion-recovery* experiment, whose pulse sequence is shown in Fig. 9.14. Initially, the magnetization is inverted by a 180° pulse, so that $M_z(0) = -M_z^0$. This inverted magnetization is then allowed to relax for a time τ . Using Eq. 9.15 with $t = \tau$ and $M_z(0) = -M_z^0$, we can see that the z -magnetization at time τ is

$$\begin{aligned} M_z(\tau) &= -2M_z^0 \exp(-R_z \tau) + M_z^0 \\ &= M_z^0 [1 - 2 \exp(-R_z \tau)]. \end{aligned}$$

After time τ a 90° pulse is applied, the resulting FID observed, and then Fourier transformed to give the spectrum. A typical set of spectra

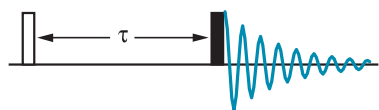


Fig. 9.14 Pulse sequence for the inversion-recovery experiment, used to estimate the value of the rate constant for longitudinal relaxation, R_z (or, alternatively, the time constant T_1).

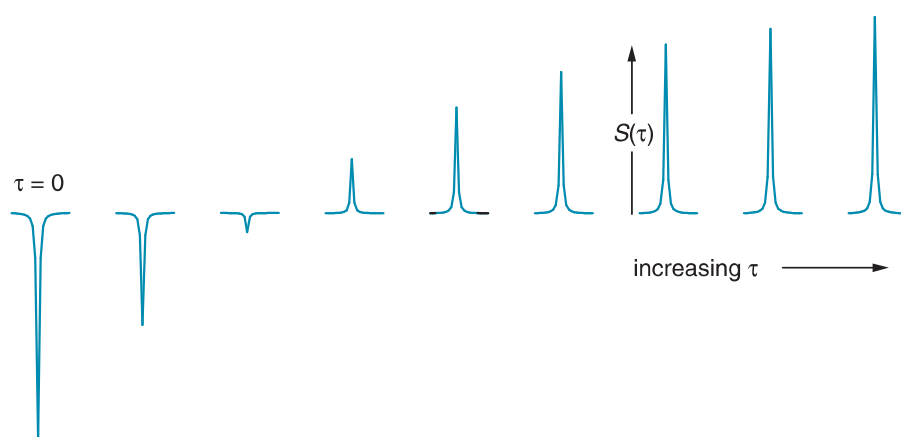


Fig. 9.15 Typical set of spectra that would be recorded using the inversion–recovery pulse sequence of Fig. 9.14 on the preceding page. The spectrum recorded for $\tau = 0$ has been phased so that the peak is negative, and then the same phase correction has been applied to all of the other spectra.

recorded for different values of τ is shown in Fig. 9.15. Note how the line starts negative, passes through zero and then becomes more positive as τ increases.

The height, $S(\tau)$, of the peak in the spectrum will be proportional to the size of the z -magnetization present just before the 90° pulse. Thus, $S(\tau)$ can be written

$$S(\tau) = c[1 - 2\exp(-R_z\tau)], \quad (9.16)$$

where c is some constant of proportion.

From Eq. 9.16, it follows that the peak height at time $\tau = 0$, $S(0)$, is $-c$. We can therefore replace the constant c in this equation by $-S(0)$ to give

$$S(\tau) = S(0)[2\exp(-R_z\tau) - 1],$$

which can be tidied up to

$$\frac{S(\tau)}{S(0)} = 2\exp(-R_z\tau) - 1.$$

This can be rearranged into a form which will give a straight-line plot in the following way:

$$\begin{aligned} \frac{S(\tau)}{S(0)} &= 2\exp(-R_z\tau) - 1 \\ \frac{S(\tau)}{S(0)} + 1 &= 2\exp(-R_z\tau) \\ \frac{S(\tau) + S(0)}{2S(0)} &= \exp(-R_z\tau) \\ \ln\left(\frac{S(\tau) + S(0)}{2S(0)}\right) &= -R_z\tau, \end{aligned}$$

where, to go to the last line, we have taken natural logarithms of both sides.

This last line implies that if we plot $\ln([S(\tau) + S(0)]/2S(0))$ against τ we will obtain a straight line of slope $-R_z$. So, all we need to do is repeat the experiment over a suitable range of τ values, measure the peak heights, and make this plot.

9.5.2 Making a quick estimate of the relaxation rate constant

Sometimes, all we want is a rough estimate of R_z , so doing a complete inversion–recovery experiment, and then plotting a graph would be rather a waste of time. One method for obtaining such a rough estimate is to use the inversion–recovery pulse sequence, but rather than varying the delay τ systematically, we just try a few values until we locate the time at which the signal goes through a null. As we will see in a moment, this null time is simply related to the value of R_z .

It is quite easy to find the null point as, for short times τ , the signal is negative, whereas for longer times it will be positive. A couple of quick experiments are therefore usually sufficient to ‘bracket’ the null point, and then a few more trial values usually enables us to home in on the precise value of the delay, τ_{null} , which gives a null.

From Eq. 9.16 on the previous page, the peak height $S(\tau)$ will be zero when

$$0 = c [1 - 2 \exp(-R_z \tau_{\text{null}})].$$

This can be rearranged to

$$\begin{aligned} \exp(-R_z \tau_{\text{null}}) &= \frac{1}{2} \\ \text{hence } R_z &= -\ln \frac{1}{2} / \tau_{\text{null}} \quad \text{or} \quad R_z = \ln 2 / \tau_{\text{null}}. \end{aligned}$$

So, simply by finding the value of τ_{null} , we can obtain an estimate of R_z . If we want T_1 instead, then the relationship is simply $T_1 = \tau_{\text{null}} / \ln 2$.

9.5.3 How long do I have to leave between experiments?

As we noted at the start of this chapter, the rate of longitudinal relaxation determines the time we have to leave between experiments in order to allow the system to come to equilibrium. We can now make some estimates, in terms of R_z (or T_1), as to how long this time actually has to be.

The first thing to realize is that the time it takes to get back to equilibrium depends on where you start from. For example, if we start out with the magnetization being inverted (i.e. along $-z$), then it will take longer to get back to equilibrium than if we start out with no z -magnetization.

Just exactly where the magnetization ends up at the end of a pulse sequence depends on the details of that sequence, so we cannot come to any general conclusions. However, assuming that there is no z -magnetization at the end of the sequence is a reasonable choice, as most sequences finish with a 90° pulse followed by data acquisition. The chances are that this pulse will rotate all of the magnetization into the transverse plane.

Let us suppose, therefore, that at the end of data acquisition there is no z -magnetization i.e. $M_z(0) = 0$. What we are going to work out is the relaxation delay, t_r , needed for the z -magnetization to return to a fraction f of its equilibrium value, that is when $M_z(t_r) = f M_z^0$. For complete return to equilibrium $f = 1$, whereas for a return to 90% of the equilibrium magnetization, $f = 0.9$.

Equation 9.14 on page 263 is a useful place to start, with $t = t_r$:

$$\ln \left(\frac{M_z(t_r) - M_z^0}{M_z(0) - M_z^0} \right) = -R_z t_r.$$

We put $M_z(0) = 0$ and $M_z(t_r) = fM_z^0$ to give

$$\ln\left(\frac{fM_z^0 - M_z^0}{-M_z^0}\right) = -R_z t_r.$$

This simplifies to

$$t_r = -\frac{1}{R_z} \ln(1 - f) \quad \text{or} \quad t_r = -T_1 \ln(1 - f).$$

What this tells us is that it takes an infinite amount of time for the spins to return completely to equilibrium ($f = 1$), a result which comes about because the rate of change of the magnetization gets slower the closer the magnetization is to equilibrium.

If, however, we lower our sights and accept 99% of the equilibrium magnetization, then the above expression predicts for $f = 0.99$ that $t_r = 4.6 \times T_1$. If 95% is acceptable, then $t_r = 3.0 \times T_1$, and for 90% $t_r = 2.3 \times T_1$. These values are illustrated in Fig. 9.16.

Therefore leaving a delay of around five times T_1 will guarantee that, to all intents and purposes, the magnetization has recovered to equilibrium. If we are more impatient, then three times T_1 will give us 95% of the magnetization – a value which is often regarded as being an acceptable compromise. Note that the time during which the FID is recorded can actually be counted into the relaxation delay t_r , as longitudinal relaxation is taking place during the FID.

Difficulties arise when not all of the spins in the sample have the same relaxation time T_1 . The most conservative approach is to set the relaxation delay according to the longest T_1 , but if this results in an unacceptably long delay, we might choose to set the relaxation delay according to an average value of T_1 .

Often it is the solvent which has the slowest relaxation, and hence the longest T_1 . Since we are not usually interested in the solvent signal, it is common to set the relaxation delay according to the behaviour of the interesting solute signals, and simply ignore the behaviour of the solvent.

9.6 Longitudinal dipolar relaxation of two spins

We now turn to a very important topic which is the relaxation behaviour of two spins which are interacting via the dipolar mechanism described in section 9.2.1 on page 249. The relaxation in this system has features which do not occur in a one-spin system, in particular it is only in a two-spin system that the phenomenon of *cross relaxation*, which gives rise to the very important nuclear Overhauser effect (NOE), is seen. We will therefore spend quite some time exploring dipolar relaxation and its consequences.

9.6.1 Energy levels and transition rates

We introduced and derived the energy levels for two coupled spins in section 3.5 on page 35. For the present discussion, it is not necessary for the spins to be coupled, but as we saw in section 3.5, the presence of the scalar

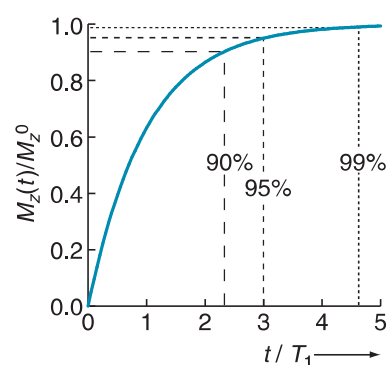


Fig. 9.16 Illustration of how long it takes to recover a certain proportion of the equilibrium z -magnetization, starting from $M_z = 0$. The horizontal time axis is expressed as a multiple of the longitudinal relaxation time, T_1 . For the magnetization to recover to 99% of its equilibrium value (i.e. $M_z/M_z^0 = 0.99$) takes almost $5 \times T_1$, but recovery to 95% only takes about $3 \times T_1$. The difference in these values arises from the fact that the rate of change of the z -magnetization gets slower the closer we are to equilibrium.

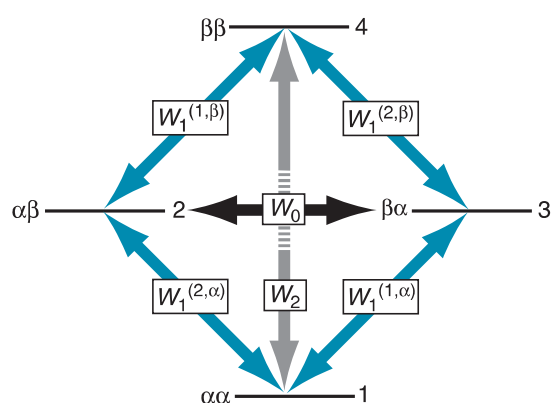


Fig. 9.17 Dipolar relaxation causes transitions between all four energy levels of a two-spin system. Four of the transitions involve a change in the total magnetic quantum number, M , by ± 1 . The corresponding rate constants are thus given a subscript ‘1’, and a superscript showing which spin is flipping and the spin state of the passive spin. So the transition from $\beta\alpha$ to $\beta\beta$ has rate constant $W_1^{(2,\beta)}$ as spin two is flipping, and spin one is in the β state. There is one double-quantum transition, with rate constant W_2 , and one zero-quantum transition, with rate constant W_0 ; as before, the subscript gives the value of ΔM . The arrangement of the energy levels shown here is appropriate for a homonuclear spin system; in a heteronuclear system the $\alpha\beta$ and $\beta\alpha$ levels will not have the same energy. However, the same set of relaxation-induced transitions occur.

coupling does not change the wavefunctions associated with the levels, but only shifts their energies very slightly.

These four energy levels can be labelled with the spin states of each spin, so the level $\alpha\alpha$ has spin one and spin two in the α state, whereas $\beta\alpha$ has spin one in the β state and spin two in the α state.

The dipolar interaction between two spins can lead to relaxation-induced transitions between *any* of these four levels, as is illustrated in Fig. 9.17. To demonstrate that this is so we would need to look at the detailed form of the Hamiltonian which describes the dipolar interaction – something which is beyond the level of this text.

Each transition has associated with it a rate constant, $W_{\Delta M}$, where the subscript gives the change, ΔM , in the total magnetic quantum number associated with the transition. Four of the transitions can be characterized as single quantum, $\Delta M = 1$, and we can further distinguish them according to which of spin one or spin two is flipping, and the spin state of spin which is not flipping (the passive spin). So, the transition from $\alpha\alpha$ to $\beta\alpha$ has rate constant $W_1^{(1,\alpha)}$ as it is spin one which is flipping, and spin two is in the α state. Similarly, the transition from $\alpha\alpha$ to $\alpha\beta$ has rate constant $W_1^{(2,\alpha)}$ as spin two is flipping, and the passive spin is in the α state.

The single-quantum relaxation rates depend on the spectral density at the frequency corresponding to the transition, which are, of course, the Larmor frequencies of the spin which is flipping. So $W_1^{(1,\alpha)}$ depends on the spectral density at the Larmor frequency of spin one, $\omega_{0,1}$, whereas $W_1^{(2,\beta)}$ depends on the spectral density at the Larmor frequency of spin two, $\omega_{0,2}$.

There is one double-quantum transition, between states $\alpha\alpha$ and $\beta\beta$, with transition rate constant W_2 . This depends on the spectral density at the

sum of the Larmor frequencies of spins one and two: $(\omega_{0,1} + \omega_{0,2})$. The zero-quantum transition rate constant, between states $\alpha\beta$ and $\beta\alpha$, is W_0 , and this depends on the spectral density at the difference of the two Larmor frequencies: $(\omega_{0,1} - \omega_{0,2})$.

9.6.2 Rate equations for the populations and z-magnetizations

Just as we did for a single spin, we can work out differential equations for the rate of change of the population of each level. Assuming that the rate is proportional to the deviation from the equilibrium population, we can write the rate of change of the population of level 1 as

$$\begin{aligned} \frac{dn_1}{dt} = & \underbrace{-W_1^{(2,\alpha)}(n_1 - n_1^0) - W_1^{(1,\alpha)}(n_1 - n_1^0) - W_2(n_1 - n_1^0)}_{\text{loss from level 1}} \\ & + \underbrace{W_1^{(2,\alpha)}(n_2 - n_2^0)}_{\text{gain from level 2}} + \underbrace{W_1^{(1,\alpha)}(n_3 - n_3^0)}_{\text{gain from level 3}} + \underbrace{W_2(n_4 - n_4^0)}_{\text{gain from level 4}}, \end{aligned}$$

where n_i is the population of the i th level, and n_i^0 is the equilibrium population of that level. The first three terms are all negative as they represent processes in which population is lost from level 1. In addition, the rates all depend on the deviation of the population of level 1 from its equilibrium value, as it is from this level that the transition is coming.

The first positive term represents a process by which the population of level 1 is increased as a result of transitions from level 2. The rate of the process therefore depends on the population of level 2. Similarly, the second and third positive terms all represent transitions in which the population of level 1 is increased.

We can write similar differential equations for each of the populations. The resulting four equations, although simple to construct, are certainly rather complex in form. However, things can be improved by rewriting the populations in terms of the z -magnetization of the two spins.

For example, following what we did before for a single spin, the z -magnetization of spin one depends on the population difference between levels 1 and 3, and between 2 and 4. Both of these population differences contribute to the spin one z -magnetization, as both transitions belong to spin one. Leaving out any constants of proportion, we can write the z -magnetization of spin one as

$$I_{1z} = (n_1 - n_3) + (n_2 - n_4).$$

We have written this as I_{1z} rather than M_{1z} in order to emphasize the connection between this z -magnetization and the operator $\hat{I}_{1z}$, which represents it in quantum mechanics.

Recognizing that transitions 1–2 and 3–4 belong to spin two, we can write

$$I_{2z} = (n_1 - n_2) + (n_3 - n_4),$$

for the z -magnetization from the second spin. It turns out that we need another magnetization term which depends on the *difference* between the population differences across the two spin-one transitions:

$$2I_{1z}I_{2z} = (n_1 - n_3) - (n_2 - n_4).$$

As with I_{1z} and I_{2z} , this term is denoted $2I_{1z}I_{2z}$ as it is related to the product operator $2\hat{I}_{1z}\hat{I}_{2z}$. In fact, simply by rearranging the populations on the right of this equation, we can see that $2I_{1z}I_{2z}$ is also the difference between the population differences across the two spin-two transitions.

$$2I_{1z}I_{2z} = (n_1 - n_2) - (n_3 - n_4).$$

$2I_{1z}I_{2z}$ is often called a ‘zz term’.

The magnetizations also have equilibrium values defined in terms of the equilibrium populations:

$$I_{1z}^0 = n_1^0 - n_3^0 + n_2^0 - n_4^0,$$

and similarly for I_{2z}^0 . It turns out that the equilibrium value of $2I_{1z}I_{2z}$ is zero.

After a lot of tedious algebra, the rate equations for the populations can be rewritten in terms of rate equations for the magnetizations:

$$\begin{aligned}\frac{dI_{1z}}{dt} &= -R_z^{(1)}(I_{1z} - I_{1z}^0) - \sigma_{12}(I_{2z} - I_{2z}^0) - \Delta^{(1)} 2I_{1z}I_{2z} \\ \frac{dI_{2z}}{dt} &= -\sigma_{12}(I_{1z} - I_{1z}^0) - R_z^{(2)}(I_{2z} - I_{2z}^0) - \Delta^{(2)} 2I_{1z}I_{2z} \\ \frac{d 2I_{1z}I_{2z}}{dt} &= -\Delta^{(1)}(I_{1z} - I_{1z}^0) - \Delta^{(2)}(I_{2z} - I_{2z}^0) - R_z^{(1,2)} 2I_{1z}I_{2z}.\end{aligned}\quad (9.17)$$

The various rate constants are defined as follows in terms of the rate constants for the individual transitions:

$$\begin{aligned}R_z^{(1)} &= W_1^{(1,\alpha)} + W_1^{(1,\beta)} + W_2 + W_0 \\ R_z^{(2)} &= W_1^{(2,\alpha)} + W_1^{(2,\beta)} + W_2 + W_0 \\ \sigma_{12} &= W_2 - W_0 \\ \Delta^{(1)} &= W_1^{(1,\alpha)} - W_1^{(1,\beta)} \\ \Delta^{(2)} &= W_1^{(2,\alpha)} - W_1^{(2,\beta)} \\ R_z^{(1,2)} &= W_1^{(1,\alpha)} + W_1^{(1,\beta)} + W_1^{(2,\alpha)} + W_1^{(2,\beta)}.\end{aligned}$$

The rate constant $R_z^{(1)}$ describes the *self relaxation* of spin one, meaning that this rate constant simply determines the rate at which the magnetization I_{1z} approaches equilibrium without the involvement of I_{2z} and $2I_{1z}I_{2z}$. Similarly, $R_z^{(2)}$ is the self-relaxation rate constant for spin two.

The rate constant σ_{12} describes the rate at which magnetization from spin one is transferred, by relaxation processes, to spin two. We make this interpretation as, in the differential equation for I_{1z} , there is a term $\sigma_{12}(I_{2z} - I_{2z}^0)$ which says that the rate of change of I_{1z} is proportional to I_{2z} , with σ_{12} as the constant of proportion. There is a similar process which transfers magnetization from spin one to spin two; it turns out to have the same rate constant.

This relaxation-mediated transfer of z -magnetization from one spin to another is called *cross relaxation*. It is an (almost) unique feature of dipolar relaxation and is, as we shall see shortly, responsible for the NOE. It is interesting to note that in scalar coupled systems we saw that it was possible

to transfer transverse anti-phase magnetization from one spin to another using RF pulses. Now we have a second mechanism for transfer, but this time it involves dipolar relaxation and z -magnetization; no scalar coupling is required.

Following the same discussion as above, $R_z^{(1,2)}$ is the rate constant for self relaxation of the $2I_{1z}I_{2z}$ term. Finally $\Delta^{(1)}$ and $\Delta^{(2)}$ are the rate constants for the interconversion of I_{1z} with $2I_{1z}I_{2z}$, and of I_{2z} with $2I_{1z}I_{2z}$. These various relaxation pathways are visualized in Fig. 9.18.

For dipolar relaxation between two spins it turns out that $W_1^{(1,\alpha)} = W_1^{(1,\beta)}$ and $W_1^{(2,\alpha)} = W_1^{(2,\beta)}$; however, in more complex situations, such as that discussed in section 9.11 on page 306, these rate constants can be different. Writing these rate constants as $W_1^{(1)}$ and $W_1^{(2)}$, respectively, the above differential equations and expressions for the rate constants simplify considerably as $\Delta^{(1)} = 0$ and $\Delta^{(2)} = 0$:

$$\begin{aligned}\frac{dI_{1z}}{dt} &= -R_z^{(1)}(I_{1z} - I_{1z}^0) - \sigma_{12}(I_{2z} - I_{2z}^0) \\ \frac{dI_{2z}}{dt} &= -\sigma_{12}(I_{1z} - I_{1z}^0) - R_z^{(2)}(I_{2z} - I_{2z}^0) \\ \frac{d2I_{1z}I_{2z}}{dt} &= -R_z^{(1,2)}2I_{1z}I_{2z}.\end{aligned}\quad (9.18)$$

The rate constants simplify to:

$$\begin{aligned}R_z^{(1)} &= 2W_1^{(1)} + W_2 + W_0 \\ R_z^{(2)} &= 2W_1^{(2)} + W_2 + W_0 \\ \sigma_{12} &= W_2 - W_0 \\ R_z^{(1,2)} &= 2W_1^{(1)} + 2W_1^{(2)}.\end{aligned}\quad (9.19)$$

We now see that there is still cross relaxation between I_{1z} and I_{2z} , but there is no relaxation-induced transfer between $2I_{1z}I_{2z}$ and either of I_{1z} or I_{2z} . Equations 9.18 are often called the *Solomon equations*; we will use them extensively in the following discussion.

9.6.3 Relaxation rate constants

The theory of how the rate constants W for transitions between individual levels are computed is beyond the scope of this text. No new quantum mechanical concepts are needed to make these calculations – it is just that the details are rather involved because we are dealing with time-dependent random processes.

What we find is that the theory predicts that the transition rate constant between two levels i and j , W_{ij} , is always the product of three terms:

$$W_{ij} = A_{ij} \times Y^2 \times j(\omega_{ij}).$$

We will consider each of these terms in turn.

A_{ij} is a number which arises from the details of the Hamiltonian which represents the particular interaction which is causing relaxation. Y^2 is a term which is related to the magnitude of the local fields which are causing

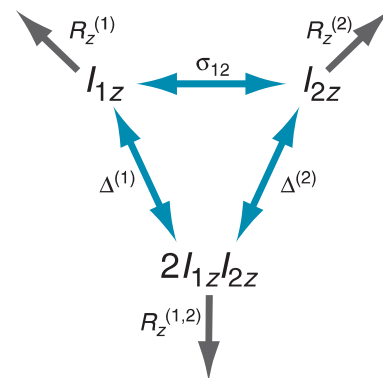


Fig. 9.18 Visualization of the relaxation pathways between different kinds of z -magnetization for two spins undergoing relaxation via the dipolar mechanism. The dark grey arrows pointing to the edges of the diagram show the self-relaxation processes i.e. those in which the magnetization simply returns to its equilibrium value. In addition, there are processes, indicated by the blue arrows, which connect the z -magnetization terms of the two spins. The most important of these is cross relaxation, with rate constant σ_{12} , which transfers magnetization between the two spins.

relaxation. The term is quite deliberately written as the square since it turns out that the rate of relaxation always depends on the average of the *square* of the local fields. Generally, Y^2 depends on the physical details of the interaction, such as the distance between two spins or the size of the chemical shift anisotropy.

Finally, $j(\omega_{ij})$ is the reduced spectral density at the frequency of the transition between the two energy levels. As we have already seen in section 9.3.2 on page 256, this is a measure of the amount of the random motion which is at the correct frequency, here ω_{ij} , needed to cause transitions between the two levels.

In the case of dipolar relaxation between two spins, the rate constants defined above are given by:

$$\begin{aligned} W_1^{(1)} &= \frac{3}{40} b^2 j(\omega_{0,1}) & W_1^{(2)} &= \frac{3}{40} b^2 j(\omega_{0,2}) \\ W_2 &= \frac{3}{10} b^2 j(\omega_{0,1} + \omega_{0,2}) & W_0 &= \frac{1}{20} b^2 j(\omega_{0,1} - \omega_{0,2}), \end{aligned}$$

where b is

$$b = \frac{\mu_0 \gamma_1 \gamma_2 \hbar}{4\pi r^3}.$$

In this expression μ_0 is a physical constant called the permeability of vacuum and which has the value $4\pi \times 10^{-7} \text{ H m}^{-1}$; ‘H’ stands for Henries, the unit of inductance. γ_1 and γ_2 are the gyromagnetic ratios of the two spins, and r is the distance between them. As described above, each rate constant consists of the product of a number, the square of a size factor and a spectral density.

Note that W_2 depends on the spectral density at the sum of the Larmor frequencies of the two spins, as this is the frequency of the transition between the $\alpha\alpha$ and $\beta\beta$ states, which are connected by W_2 . Similarly, W_0 depends on the spectral density at the difference of the two Larmor frequencies, as this is the frequency of the $\alpha\beta \leftrightarrow \beta\alpha$ transition.

Using these expressions for the various rate constants between levels, the rate constants given in Eq. 9.19 on the preceding page can be written as

$$\begin{aligned} R_z^{(1)} &= b^2 \left[\frac{3}{20} j(\omega_{0,1}) + \frac{3}{10} j(\omega_{0,1} + \omega_{0,2}) + \frac{1}{20} j(\omega_{0,1} - \omega_{0,2}) \right] \\ R_z^{(2)} &= b^2 \left[\frac{3}{20} j(\omega_{0,2}) + \frac{3}{10} j(\omega_{0,1} + \omega_{0,2}) + \frac{1}{20} j(\omega_{0,1} - \omega_{0,2}) \right] \\ \sigma_{12} &= b^2 \left[\frac{3}{10} j(\omega_{0,1} + \omega_{0,2}) - \frac{1}{20} j(\omega_{0,1} - \omega_{0,2}) \right] \\ R_z^{(1,2)} &= b^2 \left[\frac{3}{20} j(\omega_{0,1}) + \frac{3}{20} j(\omega_{0,2}) \right]. \end{aligned} \quad (9.20)$$

Note how each rate constant depends on the spectral density at more than one frequency.

9.6.4 Cross relaxation in the two motional regimes

The self-relaxation rate constants $R_z^{(1)}$ and $R_z^{(2)}$ are always positive, as they are the sum of positive terms. However, the cross-relaxation rate constant σ_{12} is the difference of two terms, and so it may be positive or negative. As we will see in this section, it turns out that the sign of σ_{12} depends on an interplay between the sizes of the Larmor frequencies and the correlation

time. We will also discover in a subsequent section that the sign of the cross-relaxation rate constant is of considerable importance in the theory of the NOE.

The situation which is of most interest is when the two spins are of the same type e.g. both protons, in which case they both have the same Larmor frequency (the tiny difference due to chemical shifts is of no significance for relaxation). In this case $j(\omega_{0,1} + \omega_{0,2})$ becomes $j(2\omega_0)$ and $j(\omega_{0,1} - \omega_{0,2})$ becomes $j(0)$. So, from Eq. 9.20 on the preceding page the cross-relaxation rate constant is

$$\sigma_{12} = \underbrace{b^2 \frac{3}{10} j(2\omega_0)}_{W_2} - \underbrace{b^2 \frac{1}{20} j(0)}_{W_0}.$$

The underbraces remind us of the origin of the two terms. It is interesting to examine the value of σ_{12} in the two motional regimes, described in section 9.3.3 on page 257.

In the fast motion limit, the reduced spectral density is simply $2\tau_c$ at all frequencies, so:

$$\begin{aligned} \text{fast motion: } \sigma_{12} &= \underbrace{b^2 \frac{3}{10} j(2\omega_0)}_{W_2} - \underbrace{b^2 \frac{1}{20} j(0)}_{W_0} \\ &= b^2 \frac{3}{10} 2\tau_c - b^2 \frac{1}{20} 2\tau_c \\ &= \frac{1}{2} b^2 \tau_c. \end{aligned}$$

From the final expression we see that in the fast motion limit σ_{12} is clearly positive. Looking back through the calculation we can see that this comes about because $W_2 > W_0$.

In the slow motion limit $j(0)$ is still $2\tau_c$, but $j(2\omega_0)$ is negligible compared with $j(0)$, so:

$$\begin{aligned} \text{slow motion: } \sigma_{12} &= \underbrace{b^2 \frac{3}{10} j(2\omega_0)}_{W_2} - \underbrace{b^2 \frac{1}{20} j(0)}_{W_0} \\ &= 0 - b^2 \frac{1}{20} 2\tau_c \\ &= -\frac{1}{10} b^2 \tau_c. \end{aligned}$$

Now, σ_{12} is negative, which we can see comes about because $W_0 > W_2$.

The cross-over point, where $\sigma_{12} = 0$, occurs when $W_2 = W_0$:

$$\text{cross-over: } \underbrace{b^2 \frac{3}{10} j(2\omega_0)}_{W_2} = \underbrace{b^2 \frac{1}{20} j(0)}_{W_0}.$$

Simply by substituting in the expression for $j(\omega)$ (Eq. 9.4 on page 257) we can see that this cross-over occurs when

$$\omega_0 \tau_c = \sqrt{\frac{5}{4}}.$$

For protons at a Larmor frequency of 500 MHz, the correlation time at this cross-over point is 360 ps, as is illustrated in Fig. 9.19. This value of the correlation time is typical of a medium-sized molecule in a more viscous solvent such as water. Small to medium-sized molecules in less

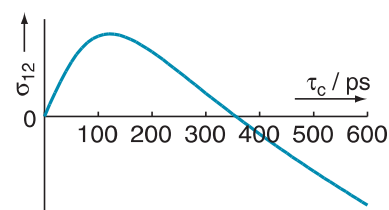


Fig. 9.19 Illustration of how the cross-relaxation rate constant, σ_{12} , changes sign from positive to negative as the correlation time is increased. The graph is computed for two protons with a Larmor frequency of 500 MHz, so the cross-over point is at $\tau_c \approx 360$ ps.

viscous solvents, such as CDCl_3 , will have smaller correlation times than this, whereas large molecules, such as proteins and polysaccharides, will have much longer correlation times than this.

In the next section we will discover that the sign of the cross-relaxation rate constant has important consequences when it comes to the NOE.

9.7 The NOE

The differential equation of the rate of change of the z -magnetization:

$$\frac{dI_{1z}}{dt} = -R_z^{(1)}(I_{1z} - I_{1z}^0) - \sigma_{12}(I_{2z} - I_{2z}^0), \quad (9.21)$$

tells us that if spin two is not at equilibrium i.e. $(I_{2z} - I_{2z}^0) \neq 0$, the rate of change of the z -magnetization on spin one will have a contribution which is proportional to the cross-relaxation rate constant, σ_{12} . This rate constant will only be non-zero if there is dipolar relaxation between the two spins, so if we find that the behaviour of the z -magnetization of spin one is affected by the amount of z -magnetization on spin two, we can deduce that cross relaxation must be taking place.

In section 9.6.3 on page 271, we saw that $\sigma_{12} \propto b^2$, so that the cross-relaxation rate goes as $1/r^6$. The rate thus falls off rather rapidly with distance, so in practice we are only likely to be able to observe the effects of cross relaxation between spins which are reasonably close to one another. In practice, for two protons, this means a distance of less than about 5 Å. Thus, if we see evidence of cross relaxation between two spins, we can be sure that they are reasonably close in space.

Cross relaxation leads to what is called the nuclear Overhauser effect (NOE), which is an exceptionally important tool when it comes to structural studies by NMR. In the following sections we will look at different ways in which cross relaxation, and hence the NOE, can be detected. All of these experiments have it in common that one spin is perturbed away from equilibrium, and then the effect of this on the z -magnetization of the other spin is determined.

9.7.1 The transient NOE experiment

The pulse sequence for the simplest transient NOE experiment is shown in Fig. 9.20. This is a difference experiment, in which the spectrum arising from sequence (b) is subtracted from that arising from sequence (a). As we shall see, taking this difference reveals the presence of the NOE in a particularly convenient way.

In sequence (a) the first thing which happens is that the z -magnetization from spin two is inverted by a selective 180° pulse (see section 4.11 on page 67); the z -magnetization from spin one is not affected by the pulse. Therefore immediately after this pulse (time $\tau = 0$) the z -magnetization of the two spins can be written as

$$\text{at } \tau = 0 : \quad I_{1z}(0) = I_{1z}^0 \quad \text{and} \quad I_{2z}(0) = -I_{2z}^0. \quad (9.22)$$

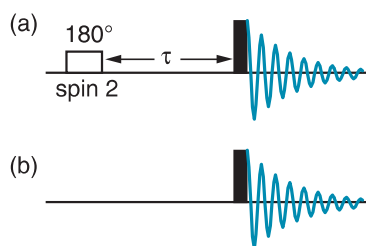


Fig. 9.20 Pulse sequence for the simple transient NOE experiment. There are two parts of the experiment. In the first part, (a), the z -magnetization of spin two is inverted by a selective 180° pulse. There then follows a delay τ during which cross relaxation takes place, and finally the z -magnetization is made observable by a non-selective 90° pulse. Experiment (a) leads to what is called the irradiated spectrum. The second experiment, (b), is simply a pulse-acquire sequence; this gives us the reference spectrum. By subtracting the reference spectrum from the irradiated spectrum, we obtain an NOE difference spectrum in which the presence of any cross relaxation to spin two is revealed.

These are the initial conditions which we would need to take into account in solving the differential equation, Eq. 9.21 on the facing page, which describes how I_{1z} varies with time.

Solving this differential equation in general terms is not fundamentally very difficult, but is a more complex task than we want to get involved in here. We are going to take a simpler approach, by solving the equation in what is called the *initial rate limit*, which only applies at short times. In this approach we assume that on the right-hand side of Eq. 9.21 I_{1z} and I_{2z} have their initial values i.e. the values at time zero given in Eq. 9.22 on the preceding page. Applying this idea gives

$$\begin{aligned} \left(\frac{dI_{1z}}{dt} \right)_{\text{init}} &= -R_z^{(1)} [I_{1z}(0) - I_{1z}^0] - \sigma_{12} [I_{2z}(0) - I_{2z}^0] \\ &= -R_z^{(1)} (I_{1z}^0 - I_{1z}^0) - \sigma_{12} (-I_{2z}^0 - I_{2z}^0) \\ &= 2\sigma_{12} I_{2z}^0. \end{aligned} \quad (9.23)$$

We have had to put the subscript ‘init’ on the differential to remind us that this expression only applies in the initial rate limit. This approximation is valid for times short enough that the z -magnetizations have not changed by very much from their initial values.

It is now easy to separate the variables (as we did in section 9.5 on page 263) in Eq. 9.23, and then integrate both sides:

$$\begin{aligned} \frac{dI_{1z}(t)}{dt} &= 2\sigma_{12} I_{2z}^0 \\ dI_{1z}(t) &= 2\sigma_{12} I_{2z}^0 dt \\ \int dI_{1z}(t) &= \int 2\sigma_{12} I_{2z}^0 dt \\ I_{1z}(t) &= 2\sigma_{12} t I_{2z}^0 + \text{const.} \end{aligned}$$

In these manipulations we have written $I_{1z}(t)$ to remind ourselves that I_{1z} depends on time. To go to the second line, we have simply taken the term dt over to the right: this separates the variables into $I_{1z}(t)$ on the left and t on the right. The two integrals are easy to compute as $\sigma_{12} I_{2z}^0$ is just a constant.

We can find the constant of integration since we know that at time zero $I_{1z}(0) = I_{1z}^0$; the constant is therefore I_{1z}^0 . Thus overall the z -magnetization of spin one at time $t = \tau$ is

$$I_{1z}(\tau) = 2\sigma_{12} \tau I_{2z}^0 + I_{1z}^0.$$

What this says is that there will be a contribution to the z -magnetization of spin one which is proportional to the time τ and the cross-relaxation rate constant σ_{12} : this is what leads to the NOE.

The next step is to work out what has happened to the z -magnetization from spin two. Again, we start with the differential equation

$$\frac{dI_{2z}}{dt} = -R_z^{(2)} (I_{2z} - I_{2z}^0) - \sigma_{12} (I_{1z} - I_{1z}^0),$$

and impose the initial conditions as before $I_{1z}(0) = I_{1z}^0$ and $I_{2z}(0) = -I_{2z}^0$ to give

$$\left(\frac{dI_{2z}}{dt} \right)_{\text{init}} = 2R_z^{(2)} I_{2z}^0.$$

This has the same form as Eq. 9.23 on the preceding page, but with a different rate constant. Integrating as before gives

$$I_{2z}(t) = 2R_z^{(2)}tI_{2z}^0 + \text{const.}$$

The initial condition is $I_{2z}(0) = -I_{2z}^0$, so the constant of integration is $-I_{2z}^0$. Thus, the z -magnetization from spin two as a function of the time τ is given by

$$I_{2z}(\tau) = 2R_z^{(2)}\tau I_{2z}^0 - I_{2z}^0.$$

Since $R_z^{(2)}\tau$ is positive, this equation says that as τ increases the initially inverted z -magnetization becomes less negative, which is the expected result as the magnetization is moving towards equilibrium.

So far we have computed the z -magnetization for each spin as a function of the time τ in pulse sequence (a) of Fig. 9.20 on page 274. For pulse sequence (b), the situation is very simple as both spins are at equilibrium just prior to the 90° pulse. The results for both experiments are summarized in the following table:

experiment	spin-one magnetization	spin-two magnetization
(a)	$I_{1z}(\tau) = 2\sigma_{12}\tau I_{2z}^0 + I_{1z}^0$	$I_{2z}(\tau) = 2R_z^{(2)}\tau I_{2z}^0 - I_{2z}^0$
(b)	I_{1z}^0	I_{2z}^0

The 90° pulse in both experiments rotates any z -magnetization into the transverse plane, where it is observed as an FID. Fourier transformation of the FID will give us a spectrum in which there is one peak at the offset (shift) of spin one, and similarly a peak at the offset of spin two.

The height of the peak for spin one, $S_1(\tau)$, will be proportional to $I_{1z}(\tau)$, and similarly the height of the spin-two peak, $S_2(\tau)$, will be proportional to $I_{2z}(\tau)$. Furthermore, if both spins are of the same type (e.g. proton), their equilibrium z -magnetizations will be equal: $I_{1z}^0 = I_{2z}^0$. Thus, the data in the table above can be converted into peak heights (in the table, c is the constant of proportion):

spectrum	$S_1(\tau)$	$S_2(\tau)$
irradiated: (a)	$c(1 + 2\sigma_{12}\tau)$	$c(2R_z^{(2)}\tau - 1)$
reference: (b)	c	c
NOE difference: (a) - (b)	$c2\sigma_{12}\tau$	$c(2R_z^{(2)}\tau - 2)$

Spectrum (a) is called the *irradiated* spectrum, as it is from the experiment in which one of the spins was inverted. Spectrum (b) is called the *reference* spectrum; it is simply the normal spectrum. The table also shows the peak heights in what is known as the *NOE difference* spectrum, found by taking (a) - (b).

To interpret these results we will assume, for the sake of argument, that σ_{12} is positive. We must also remember that these results are only valid in the initial rate limit, which in practice means that $\sigma_{12}\tau \ll 1$ and $R_z^{(2)}\tau \ll 1$.

In spectrum (a), the peak for spin one is, as a result of cross relaxation, a little higher than the corresponding peak in the reference experiment, (b). We say that the peak has received an *NOE enhancement* or, more loosely, that the peak ‘has an NOE’. The presence of this enhancement is revealed by subtracting the reference spectrum (b) from the irradiated spectrum (a), as doing so just leaves the intensity which arose due to cross relaxation i.e. the NOE enhancement. The difference (a) – (b) is called the *NOE difference spectrum*.

The peak for spin two in spectrum (a) is negative, on account of the fact that the spin was inverted, with a peak height which is very close to minus the peak height in the reference experiment (b). Thus, in the NOE difference spectrum, the spin-two peak is negative, with a peak height of more or less twice that in the reference spectrum. The process of computing the NOE difference spectrum is shown in Fig. 9.21.

If there is no cross relaxation i.e. $\sigma_{12} = 0$, then the intensity of the spin-one peak in the irradiated spectrum (a) will be the same as in the reference spectrum (b), and so the spin-one peak will not appear in the NOE difference spectrum. This difference spectrum is therefore a very direct way of seeing which spins are cross relaxing with the spin which is inverted (in this case, spin two). In this spectrum, the only peaks we will see are those which receive an NOE enhancement and the one which was initially irradiated.

In practice, it is rather difficult to record high-quality NOE difference spectra using the simple pulse sequence described in this section. However, by replacing the selective 180° pulse with an alternative inversion sequence which uses pulsed field gradients, it is possible to obtain excellent spectra on a routine basis. The details of this modified experiment are given in section 11.16 on page 432.

The NOE enhancement

The size of the NOE enhancement, η , is expressed as a fraction in the following way:

$$\eta = \frac{\text{peak height in irradiated spectrum} - \text{peak height in reference spectrum}}{\text{peak height in reference spectrum}}.$$

For the above example the enhancement is computed as

$$\begin{aligned}\eta &= \frac{c(1 + 2\sigma_{12}\tau) - c}{c} \\ &= 2\sigma_{12}\tau.\end{aligned}$$

For molecules in the fast motion limit, σ_{12} is positive and so is the enhancement. In the slow motion limit, σ_{12} , and therefore the enhancement, is negative.

The larger the cross-relaxation rate constant σ_{12} , the greater the enhancement. Recall from section 9.6.3 on page 271 that $\sigma_{12} \propto r^{-6}$, so spins which are closer will have faster cross relaxation, and hence show larger enhancements. It is possible, therefore, to use the size of the enhancement as an indicator of the internuclear distance.

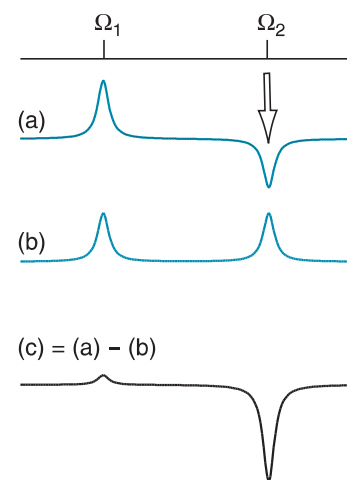


Fig. 9.21 Illustration of how NOE difference spectra are constructed. Spectrum (a) is the irradiated spectrum, recorded using sequence (a) of Fig. 9.20 on page 274. Spin two has been inverted, as indicated by the arrow and the negative intensity of the corresponding peak. In fact, in this spectrum the peak from spin one is slightly higher than in the reference spectrum, (b), on account of there being cross relaxation from spin two. This increase in peak height is most simply visualized by computing the NOE difference spectrum, (c), as (a) – (b). Now we can clearly see that spin one has received an NOE enhancement. In the difference spectrum, the irradiated peak appears with negative intensity. It has been assumed that the cross-relaxation rate constant is positive, so the NOE enhancement is positive.

To use the NOE enhancement as a *quantitative* measure of the distance turns out to be rather difficult due to a number of theoretical and practical limitations. To find out more about this, you should refer to the excellent text by Neuhaus and Williamson (see Further reading).

Longer mixing times

Remember that all of our calculations so far are in the initial rate limit. To work out what happens at longer times, we need to integrate the Solomon equations without making the restrictive assumptions we used above. We will not go into the details here but just describe the outcome.

To start with, the NOE enhancement increases linearly – this is what we predicted using the initial rate approximation. However, at longer times, the rate of increase of the enhancement starts to slow down, and eventually it reaches a maximum. After this, the enhancement steadily falls to zero.

The maximum value of the enhancement, and the time at which this occurs, is a function of the cross-relaxation and self-relaxation rate constants. Not surprisingly, the greater σ_{12} , the greater the maximum and the earlier time at which it occurs. The effect of increasing the self-relaxation rate constants is to decrease the maximum enhancement.

9.7.2 The steady-state NOE experiment

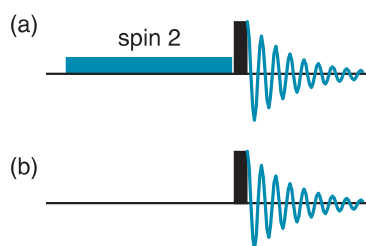


Fig. 9.22 Pulse sequence used to record a steady-state NOE difference spectrum. Two experiments are needed: in (a) the target spin (here spin two) is irradiated with a weak field so as to saturate it. The irradiation is applied for long enough for a new steady state to be reached. Experiment (b) is just a normal pulse–acquire sequence, which leads to the reference spectrum. The NOE difference spectrum is found by taking (a) – (b).

In this experiment, rather than inverting one of the spins, and then letting cross relaxation take place, the target spin is irradiated continuously with a weak RF field. The field is chosen to be weak enough that only the spin with which it is on-resonance is affected.

The result of this irradiation is to *saturate* the target spin, which means that its z -magnetization is forced to zero. The term saturation comes about from thinking about the populations of the two energy levels. Continuous irradiation eventually equalizes these populations, a situation which in spectroscopy is described as saturation. The population difference, and hence the z -magnetization, thus goes to zero.

The pulse sequence used to measure a steady-state NOE is shown in Fig. 9.22. As for the transient experiment, two spectra are recorded: one in which the target spin (here spin two) is saturated, and a reference spectrum in which all spins are at equilibrium.

As before, we can analyse the experiment using the Solomon equations. However, the approach is a little different to that used for the transient experiment. There are two key ideas needed here. The first is that as spin two is irradiated continuously it is kept saturated, and so we can assume that there is no spin-two magnetization i.e. $I_{2z} = 0$ at all times. The second is that if we saturate spin two for long enough we will reach a new steady state in which the spin-one magnetization is not changing with time, i.e.

$$\left(\frac{dI_{1z}}{dt} \right)_{ss} = 0;$$

the subscript ‘SS’ indicates the new steady state. As before we start with the Solomon equation for the spin-one magnetization (Eq. 9.21 on page 274):

$$\frac{dI_{1z}}{dt} = -R_z^{(1)}(I_{1z} - I_{1z}^0) - \sigma_{12}(I_{2z} - I_{2z}^0).$$

Applying the two conditions we have described gives

$$0 = -R_z^{(1)}(I_{1z,SS} - I_{1z}^0) - \sigma_{12}(-I_{2z}^0),$$

where $I_{1z,SS}$ is the steady-state value of the z -magnetization on spin one. Rearranging this last expression gives

$$I_{1z,SS} = \frac{\sigma_{12}}{R_z^{(1)}} I_{2z}^0 + I_{1z}^0.$$

As with the transient experiment, the z -magnetization on spin one is altered as a result of cross relaxation.

If we assume, as we did before, that $I_{1z}^0 = I_{2z}^0$, then the peak height of the spin-one resonance in the steady-state spectrum is $c(1 + \sigma_{12}/R_z^{(1)})$. Since spin two is saturated, no spin-two peak appears in the irradiated experiment. In the reference spectrum, both peaks have height c . Therefore in the NOE difference spectrum the spin-one peak will have height $c\sigma_{12}/R_z^{(1)}$, and the spin-two peak will have height $-c$; this is illustrated in Fig. 9.23. As before, the difference spectrum reveals the presence of cross relaxation in a very convenient way.

The NOE enhancement is given by

$$\begin{aligned} \eta_{SS} &= \frac{c(1 + \sigma_{12}/R_z^{(1)}) - c}{c} \\ &= \frac{\sigma_{12}}{R_z^{(1)}}. \end{aligned}$$

As in the transient experiment, an NOE enhancement will only be seen if the cross-relaxation rate constant is non-zero. However, in contrast to the transient experiment, the steady-state NOE depends not just on σ_{12} , but the ratio of this cross-relaxation rate constant to the self-relaxation rate constant of spin one (i.e. the spin which is enhanced).

Thus the size of the steady-state NOE does not simply reflect the amount of cross relaxation, but rather the balance between cross and self relaxation. Cross relaxation must be from the dipolar mechanism, but self relaxation can be from this and other sources (such as dissolved oxygen). We therefore have no way of relating the self-relaxation rate constant, and hence the NOE enhancement, to the internuclear distance. Steady-state NOE measurements can therefore only be used qualitatively.

The final point is to estimate how long it takes the spins to come to the new steady state once we start irradiating one of them. As we have seen, the steady-state enhancement depends on a balance between the self and cross relaxation of spin one, so the rate at which the system comes to a steady state clearly has something to do with these rate constants. It turns out that the smaller of these is always σ_{12} , so the value of this rate constant is the limiting factor in determining the rate at which the steady state is achieved.

For a first-order process, the reciprocal of the rate constant gives a related parameter known as the *time constant*. It can be shown that

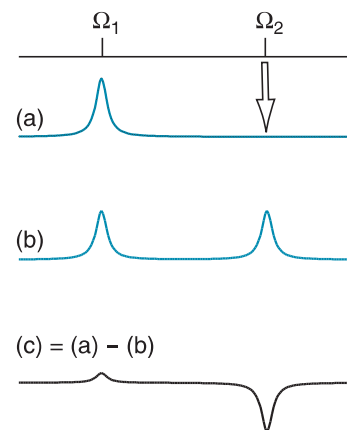


Fig. 9.23 Illustration of how a steady-state NOE difference spectrum is constructed. The spectra are similar to those of Fig. 9.21 on page 277 except that spin two is saturated, so does not appear in the irradiated spectrum (a). Subtracting the reference spectrum (b) from the irradiated spectrum (a) gives the NOE difference spectrum (c). As before, the peak from the irradiated spin is negative, and the enhancement is visible on spin one. It has been assumed that the cross-relaxation rate constant is positive.

equilibrium is only reached after waiting several times this time constant. So, in the case of the steady-state NOE, we need to wait several times $1/\sigma_{12}$. This can easily be several seconds. However, since there is little quantitative information that we can derive from the size of the steady-state NOE enhancements, it is not necessary to wait until the steady state has been reached.

9.7.3 Heteronuclear steady-state NOE

An important application of the steady-state NOE is to enhance the intensity of signals recorded from heteronuclei such as ^{13}C . The idea is that by irradiating the protons, the z -magnetization of the ^{13}C nuclei will be enhanced as a result of cross relaxation. Thus, when the z -magnetization is rotated into the transverse plane, stronger ^{13}C signals will be observed as a result of the NOE enhancement.

Since this is a heteronuclear experiment, we will switch to the IS notation, where the I spins are typically protons. Adapting the result of the previous section, the enhanced z -magnetization on the S spin (^{13}C) as a result of saturating the I spin (proton) is:

$$S_{z,\text{ss}} = \frac{\sigma_{IS}}{R_z^{(S)}} I_z^0 + S_z^0. \quad (9.24)$$

In section 9.6.2 on page 269 we defined I_z in terms of the population difference across the I spin transitions, so I_z^0 depends on the equilibrium population difference. It was shown in section 9.4.1 on page 259 that this difference is proportional to the gyromagnetic ratio of the spin, and so it follows that

$$\frac{S_z^0}{I_z^0} = \frac{\gamma_S}{\gamma_I}. \quad (9.25)$$

Therefore as γ for proton is about four times γ for ^{13}C , the equilibrium magnetization of proton is four times that of ^{13}C .

If we use Eq. 9.25 to write I_z^0 in terms of S_z^0 in Eq. 9.24, we find that the steady-state magnetization on the S spin is

$$S_{z,\text{ss}} = \frac{\gamma_I \sigma_{IS}}{\gamma_S R_z^{(S)}} S_z^0 + S_z^0.$$

Using this we can determine that the NOE enhancement of the S spin is

$$\eta_{\text{ss}} = \frac{\gamma_I \sigma_{IS}}{\gamma_S R_z^{(S)}}.$$

An estimate for this enhancement can be made if it is assumed that, for a ^{13}C with a directly attached proton, the self relaxation of the ^{13}C is dominated by its dipolar interaction with the proton. We will also assume that we are in the fast motion limit, so that all spectral densities are simply $2\tau_c$.

With these assumptions, the relevant expressions for the rate constants given in Eq. 9.20 on page 272 simplify to:

$$R_z^{(S)} = b^2 \tau_c \quad \sigma_{IS} = \frac{1}{2} b^2 \tau_c.$$

Substituting these into the expression for η_{SS} we find

$$\text{fast motion: } \eta_{SS} = \frac{\gamma_I}{2\gamma_S},$$

which, in the case of spin I being proton and spin S being ^{13}C , gives $\eta_{SS} \approx 2$. Clearly, there is a substantial enhancement of the ^{13}C signal to be obtained in this way.

In practice, the enhancement we obtain will be less than this estimate. There are two main reasons for this. First, there are probably other sources of relaxation of the S spin (the ^{13}C) than the attached proton. As a result, $R_z^{(S)}$ is increased and the enhancement is reduced. Secondly, we rarely have the patience to wait the rather long time which is needed for the steady state to be reached. Nevertheless, useful enhancements can be achieved with somewhat shorter times.

9.7.4 Two-dimensional NOESY

A two-dimensional NOESY spectrum looks very similar to a COSY spectrum, with the important exception that the cross peaks are generated, not by coherence transfer through couplings, but by cross relaxation. Thus, the appearance of a NOESY cross peak at $\{\Omega_i, \Omega_j\}$ tells us that there is cross relaxation between spins i and j – in other words, the two spins must be reasonably close in space. COSY and NOESY are therefore complementary experiments: the former tells us which spins are coupled, and so connected by the bonding network, whereas the latter tells us which spins are close in space.

The NOESY pulse sequence is shown in Fig. 9.24. We will analyse it for the case of two spins which are undergoing dipolar relaxation and, for simplicity, we will assume that there is no scalar coupling between the two spins.

The first part of the sequence, $90^\circ - t_1 - 90^\circ$, has already been analysed in detail when discussing the COSY experiment. We are only interested in the z -magnetization present after the second pulse, as it is such magnetization which can undergo transfer due to cross relaxation. Therefore we need term [1] from page 191:

$$-\cos(\pi J_{12}t_1) \cos(\Omega_1 t_1) \hat{I}_{1z}. \quad [1]$$

As we are assuming that there is no coupling, $J_{12} = 0$ and hence $\cos(\pi J_{12}t_1) = 1$, giving

$$-\cos(\Omega_1 t_1) \hat{I}_{1z}.$$

This term arises from the equilibrium magnetization on spin one; there is an analogous term arising from the spin-two equilibrium magnetization:

$$-\cos(\Omega_2 t_1) \hat{I}_{2z}.$$

Therefore, just after the second pulse ($\tau = 0$) the z -magnetizations on the two spins are:

$$I_{1z} = -\cos(\Omega_1 t_1) I_{1z}^0 \quad \text{and} \quad I_{2z} = -\cos(\Omega_2 t_1) I_{2z}^0.$$

NOESY: Nuclear Overhauser Effect Spectroscopy

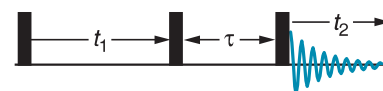


Fig. 9.24 The NOESY pulse sequence. During t_1 , transverse magnetization acquires a phase label according to the offset; this transverse magnetization is rotated onto the z -axis by the second pulse. During the mixing time τ cross relaxation may transfer this labelled z -magnetization to other spins. The final pulse rotates the z -magnetization into the transverse plane, allowing a signal to be detected.

Essentially what we have here are z -magnetizations which carry a label with them, in the form of the modulation $\cos(\Omega_i t_1)$, identifying them as being from spin one or spin two. During the mixing time, these magnetizations may be transferred to another spin, carrying with them the label which identifies their source. We will see that such transfers are the origin of the cross peaks in the spectrum.

We are going to assume that the two spins are of the same type, so that their equilibrium z -magnetizations are equal; we will write these as I_z^0 . Also, to save space we will use the notation

$$c_1 = -\cos(\Omega_1 t_1) \quad \text{and} \quad c_2 = -\cos(\Omega_2 t_1).$$

Further, as there is only one cross-relaxation rate constant, we will write it as σ , and we will also assume that the two self-relaxation rate constants are equal, with value R_z .

With all of these simplifications, the Solomon equations (Eq. 9.18 on page 271) become

$$\begin{aligned} \frac{dI_{1z}(t)}{dt} &= -R_z [I_{1z}(t) - I_z^0] - \sigma [I_{2z}(t) - I_z^0] \\ \frac{dI_{2z}(t)}{dt} &= -\sigma [I_{1z}(t) - I_z^0] - R_z [I_{2z}(t) - I_z^0]. \end{aligned} \quad (9.26)$$

We are going to solve these using the initial rate approximation, just as we did for the transient NOE experiment in section 9.7.1 on page 274. The initial conditions are

$$I_{1z}(0) = c_1 I_z^0 \quad \text{and} \quad I_{2z}(0) = c_2 I_z^0. \quad (9.27)$$

Inserting these on the right-hand sides of Eq. 9.26 gives, in the initial rate limit,

$$\begin{aligned} \left(\frac{dI_{1z}(t)}{dt} \right)_{\text{init}} &= -R_z(c_1 - 1)I_z^0 - \sigma(c_2 - 1)I_z^0 \\ \left(\frac{dI_{2z}(t)}{dt} \right)_{\text{init}} &= -\sigma(c_1 - 1)I_z^0 - R_z(c_2 - 1)I_z^0. \end{aligned}$$

These equations can be integrated, just as we did before, to give

$$\begin{aligned} I_{1z}(t) &= -R_z t(c_1 - 1)I_z^0 - \sigma t(c_2 - 1)I_z^0 + \text{const. 1} \\ I_{2z}(t) &= -\sigma t(c_1 - 1)I_z^0 - R_z t(c_2 - 1)I_z^0 + \text{const. 2} \end{aligned}$$

The initial conditions given in Eq. 9.27 enable us to find the constants as:

$$\text{const. 1} = c_1 I_z^0 \quad \text{and} \quad \text{const. 2} = c_2 I_z^0.$$

Putting these constants back into the integrated equations, tidying up and finally setting the time to τ (the mixing time) gives the following expressions for the z -magnetizations at the end of the mixing time:

$$\frac{I_{1z}(\tau)}{I_z^0} = c_1(1 - R_z \tau) - c_2 \sigma \tau + (R_z + \sigma) \tau \quad (9.28)$$

$$\frac{I_{2z}(\tau)}{I_z^0} = c_2(1 - R_z \tau) - c_1 \sigma \tau + (R_z + \sigma) \tau. \quad (9.29)$$

The final pulse in the sequence rotates this z -magnetization into the transverse plane, where it is then observed during t_2 . Without any detailed calculations we can see that the spin-one z -magnetization in Eq. 9.28 on the facing page will give rise to peaks at the offset of spin one, Ω_1 , in the ω_2 dimension. From the right-hand side of Eq. 9.28 on the preceding page, we can see that there are three terms modulating this peak in t_1 . Writing out c_1 and c_2 in full, these terms are:

$$\underbrace{-\cos(\Omega_1 t_1)(1 - R_z \tau)}_{\text{diagonal}} + \underbrace{\cos(\Omega_2 t_1) \sigma \tau}_{\text{cross}} + \underbrace{(R_z + \sigma) \tau}_{\text{axial}}.$$

The first term is modulated at Ω_1 in t_1 and is therefore a diagonal peak, which will appear at $\{\omega_1, \omega_2\} = \{\Omega_1, \Omega_1\}$; its intensity is $(R_z \tau - 1)$. Recall that in this initial rate limit $R_z \tau \ll 1$, so this peak is negative. Our interpretation of the diagonal peak is that it arises from z -magnetization which started out on spin one and remained on that spin during the mixing time.

The second term is modulated at Ω_2 in t_1 , and so gives rise to a cross peak at $\{\Omega_2, \Omega_1\}$. The intensity of this peak is $(\sigma \tau)$, which means that if σ is positive, it will be small and positive i.e. of *opposite* sign to the diagonal. On the other hand, if we are in the slow motion limit, which gives $\sigma < 0$, the cross peak will have the *same* sign as the diagonal. This term arises from magnetization which started on spin two at the beginning of the mixing time and was then transferred to spin one as a result of cross relaxation. If there is no cross relaxation ($\sigma = 0$), then there are no cross peaks.

Finally, the third term has no modulation in t_1 , and so will appear at $\omega_1 = 0$; such peaks are called *axial peaks*. The magnetization responsible for the axial peak has lost its frequency label as this is the magnetization which has recovered as a result of relaxation during the mixing time. The intensity of the axial peaks goes as $(R_z + \sigma) \tau$, and as it turns out that $R_z > |\sigma|$, these peaks will be positive, i.e. opposite in sign to the diagonal. Figure 9.25 illustrates the way in which all three types of peaks appear in the spectrum.

From Eq. 9.29 on the preceding page we can see that there is a complementary set of peaks arising from the z -magnetization on spin two present at the end of the mixing time: a diagonal peak at $\{\Omega_2, \Omega_2\}$, a cross peak at $\{\Omega_1, \Omega_2\}$, and an axial peak at $\{0, \Omega_2\}$. The intensities of these peaks are the same as their counterparts appearing at Ω_1 in ω_2 .

In many ways, NOESY is the two-dimensional counterpart of the transient NOE experiment described in section 9.7.1 on page 274. As in that experiment, at first increasing τ increases the intensity of the cross peaks. Then, the rate of increase decreases until we reach a maximum, after which the intensity falls off.

Suppression of axial peaks

The axial peaks convey no useful information and can be somewhat troublesome if they obscure the wanted cross peaks. It would be a good idea, therefore, to have a method of suppressing them.

Looking back over the calculation, you can see that the distinguishing feature of the terms which give rise to the axial peaks is that they have no modulation as a function of t_1 . The reason why this is so is that the magnetization which gives rise to the axial peaks is created as a result of relaxation during τ .

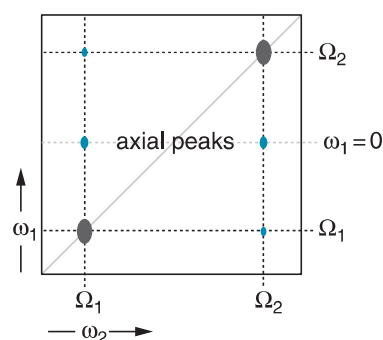
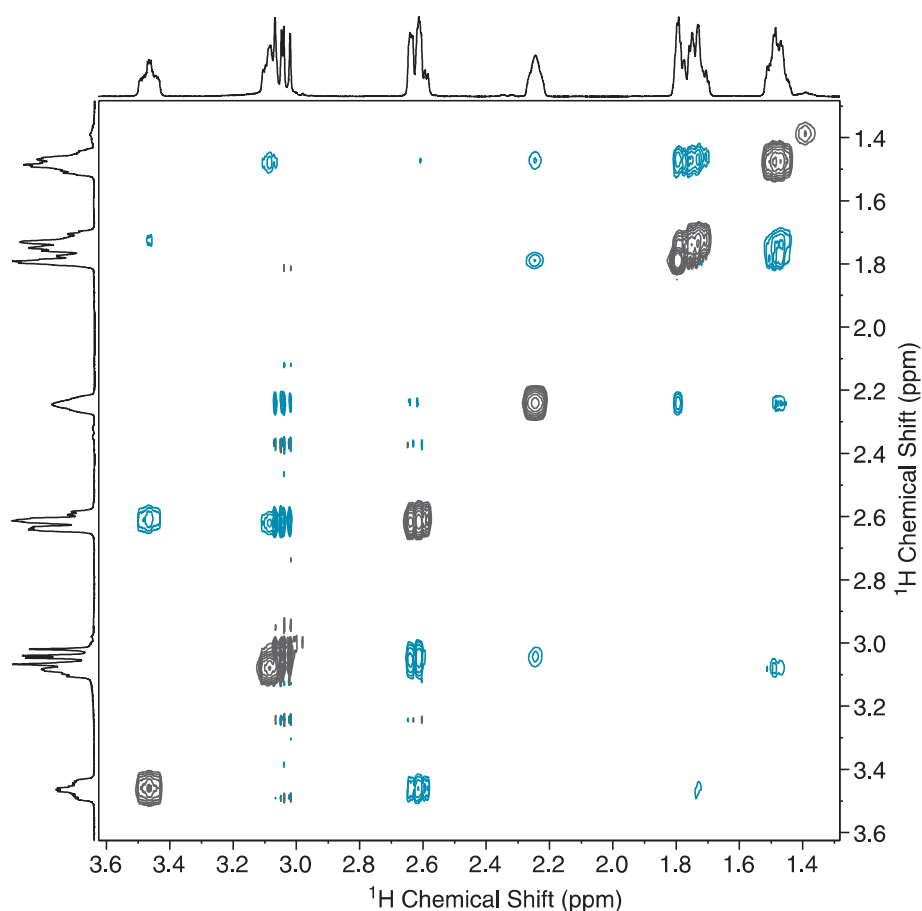


Fig. 9.25 Schematic NOESY spectrum for two spins undergoing cross relaxation. Positive peaks are shown in blue and negative in dark grey; the spectrum is shown for a molecule in the fast motion limit i.e. $\sigma > 0$. The spectrum is closely analogous to a COSY, except that in NOESY the cross peaks arise due to cross relaxation during the mixing time. In addition to the diagonal peaks and cross peaks, the spectrum shows axial peaks which appear at $\omega_1 = 0$.

Fig. 9.26 Part of the NOESY spectrum of quinine, recorded at 500 MHz and with a mixing time of 1 s. As expected for this small molecule, the cross and diagonal peaks have opposite signs. The region shown is the same as for the DQF COSY spectrum shown in Fig. 8.17 on page 202. The contribution from zero-quantum coherence present during the mixing time has been suppressed using the method described in section 11.15 on page 426.



If the phase of the first pulse in the sequence is changed from x to $-x$, then the sign of the z -magnetization at the start of the mixing time is also changed. This sign change propagates through the subsequent calculations, and results in both the diagonal and cross peaks changing sign. However, the axial peaks do not change sign as they arise from recovered magnetization.

So, we can use a simple difference method to suppress the axial peaks. The experiment is repeated twice, once with the phase of the first pulse set to x and one with the phase set to $-x$. Subtracting the two experiments results in the cancellation of the axial peaks, while the cross and diagonal peaks add up.

Figure 9.26 shows part of the NOESY spectrum of quinine. For this small molecule the cross-relaxation rate constant is positive, and so the diagonal and cross peaks have opposite sign. Some of the cross peaks are quite strong, whereas others are much weaker, implying that they are between more distant spins.

9.7.5 The NOE in more extended spin systems

The dipolar interaction involves a pair of spins, so even in an extended spin system we can still think of the cross relaxation between two particular spins, and hence the NOE which will be seen between them. The presence of other spins does affect the NOE, however, as they provide additional sources of relaxation. An NOE is always a competition between the

transfer due to cross relaxation and the general loss of magnetization caused by self relaxation of the two spins involved. If the self relaxation is fast enough, then it will outcompete the cross relaxation, and no NOE will be seen. It is essentially this competition which sets a limit on the maximum distance over which we are able to observe the effects of cross relaxation.

In an extended spin system it is possible that NOE enhancements between spins which are *not* cross relaxing one another will be observed. This comes about in the following way. Suppose we have three spins, A, B and C. A is close to B, so there is cross relaxation between them. Similarly, B and C are close to one another, and so cross relax. However, there is no cross relaxation between A and C; the situation is depicted in Fig. 9.27.

Let us assume that we are in the fast motion limit, so σ is positive. If we selectively invert A in a transient NOE experiment, then we will see a positive NOE enhancement on B. This enhancement means that the z -magnetization on spin B has been made *greater* than its equilibrium value.

Consequently, as spin B is no longer at equilibrium, cross relaxation with spin C will cause magnetization from B to be transferred to C. As a result, an NOE enhancement will be seen on spin C. Since the magnetization on B is *greater* than the equilibrium value, cross relaxation to C results in a negative enhancement of that spin, even though σ is positive. Figure 9.27 shows the resulting NOE difference experiment.

The overall result is that inversion of A results in a positive NOE enhancement of B, and a *negative* NOE enhancement on the remote spin C, even though this latter spin is not cross relaxing with A. Spin C is said to receive a *relayed* NOE from A. It turns out that in the slow motion limit, where σ is negative, both the direct and relayed NOE enhancements are negative.

Such relayed NOEs are potentially confusing as they break the simple interpretation that observation of an NOE between two spins means that they must be close in space. However, in the fast motion limit, the sign of the NOE enhancement does give us a way of distinguishing the direct from the relayed NOEs. In the slow motion limit, all the NOEs are negative, and therefore cannot be distinguished by their signs.

The final point is that these relayed NOEs generally build up much more slowly than the direct NOEs. This is because the relayed NOE requires first that a normal direct NOE be generated, and then that this enhancement causes a second NOE to the remote spin. By restricting ourselves to modest mixing times, it is rather unlikely that such transferred NOEs will have had time to build up.

In the slow motion limit, the cross-relaxation rate constant depends only on W_0 , which in turn depends on $j(0)$, where $j(0) = 2\tau_c$. As the correlation time gets longer and longer, the cross-relaxation rate therefore also increases. So, for large molecules with long correlation times, such as proteins, cross relaxation can be quite efficient. In such cases, relayed NOE enhancements can build up quickly, and indeed multiple relays along a chain of spins are also possible. This phenomenon, in which magnetization is spread amongst the spins by efficient cross relaxation, is called *spin diffusion*. The presence of this effect leads to ambiguities in the interpretation of the observed NOE enhancements.

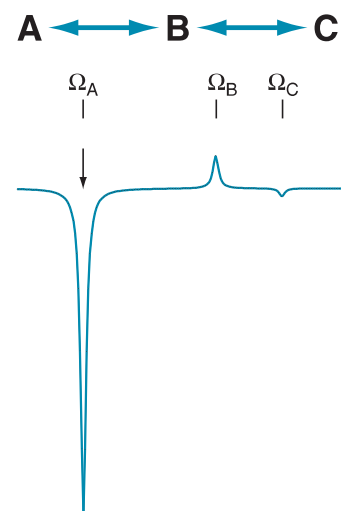


Fig. 9.27 Schematic NOE difference spectrum for three spins, A, B and C, in a line. Cross relaxation between the spins is indicated by the double-headed arrows; note that there is no cross relaxation between A and C. The lower part of the figure shows the NOE difference spectrum expected for the case where A is irradiated. As it is assumed that the cross-relaxation rate constant is positive, we see a positive NOE enhancement on B. Even though A and C are not cross relaxing one another, we see a negative enhancement on C. This is due to a two-stage transfer, first from A to B, and then from B to C. The NOE on C is described as a relayed NOE. Note that for the purposes of the diagram, the size of the relayed NOE has been greatly exaggerated.

9.8 Transverse relaxation

We now turn our attention to a more detailed discussion of transverse relaxation. In section 9.1.4 on page 247 it was noted that there are two contributions to transverse relaxation: the non-secular part, which arises from the same fluctuations in the local magnetic fields which give rise to longitudinal relaxation, and the secular part, which arises from the z -components of these fields. Unsurprisingly, the rates of relaxation due to both of these contributions can be described in a similar way to that we have already used for longitudinal relaxation, with the spectral density and the correlation time playing a central role. However, there are some subtle differences between the secular and non-secular contributions, and so we first turn to a discussion of these. A useful way of understanding the origin of the secular contribution is to draw an analogy between it and the process of chemical exchange, which is therefore the topic of the next section.

9.8.1 Chemical exchange

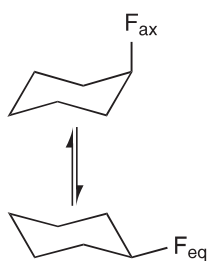


Fig. 9.28 In fluorocyclohexane the fluorine can be either axial or equatorial, and these two conformers are in dynamic equilibrium with one another. F_{ax} and F_{eq} will have different chemical shifts, but if the exchange rate between the two conformers is fast compared with the frequency separation between the two fluorine resonances a single line is seen in the spectrum: this is called *fast exchange*. If the exchange rate is slow compared with the frequency separation, then separate lines are seen for the two fluorine environments: this is called *slow exchange*.

Chemical exchange is exemplified by the simple equilibrium illustrated in Fig. 9.28. The fluorine in the axial position is expected to have a different chemical shift from that in the equatorial position. However, what we see in the spectrum of this molecule depends on the comparison between the rate constant for the exchange process and the frequency separation between the resonances corresponding to the two fluorine environments.

If the rate constant for exchange is much less than the frequency separation we expect in the (proton-decoupled) ^{19}F spectrum to see two lines, one from F_{ax} and one from F_{eq} . However, if the rate constant is much greater than the frequency separation we expect to see just one line at a position somewhere between the shifts of F_{ax} and F_{eq} . The former situation is called *slow exchange*, whereas the latter is called *fast exchange*.

Figure 9.29 on the facing page is a simulation showing how the spectra for a simple $A \rightleftharpoons B$ equilibrium change as the rate constant for the exchange process is increased. In the simulation it is assumed that the equilibrium populations of A and B are the same, so the rate constants for the forward and back processes are the same (k_{ex}). When there is no exchange ($k_{ex} = 0 \text{ s}^{-1}$, shown on the bottom left) we see two lines, one at the shift of A and one at the shift of B . The width of these lines is determined by the rate constant for transverse relaxation, which has simply been set to some convenient value.

The spectra on the left-hand side of the figure show that as the exchange rate constant increases the two lines first begin to broaden, then overlap, and eventually merge to give a single line. The theory indicates that this coalescence of the two lines occurs when the exchange rate constant is 2.22 times the frequency separation of A and B . In this case the separation is 160 Hz and so coalescence occurs when $k_{ex} = 350 \text{ s}^{-1}$.

Further increases in the exchange rate constant cause this single merged line to become narrower, as is illustrated in the spectra on the right-hand side of Fig. 9.29. Eventually, when the exchange rate constant becomes very much larger than the frequency separation, the line has the same width as it did in the absence of exchange. The position of the merged line depends on

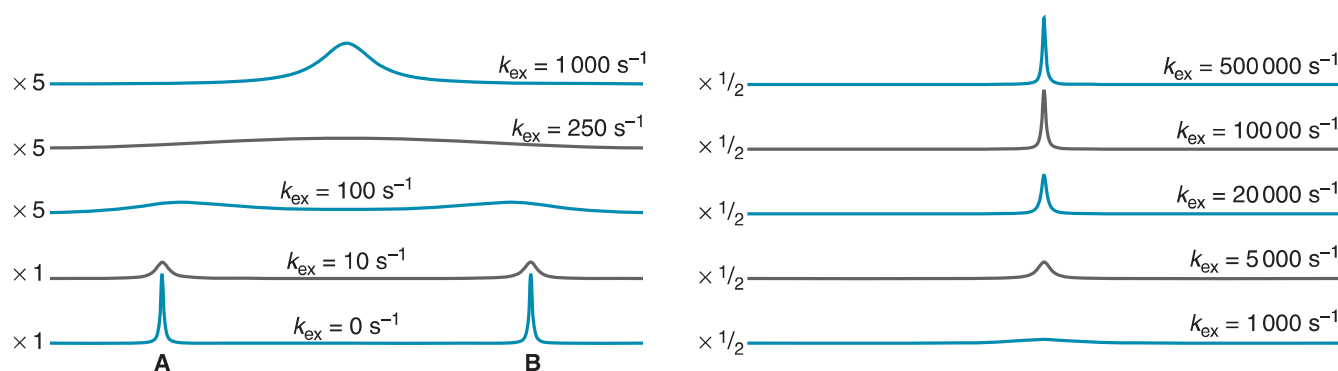


Fig. 9.29 Simulated spectra showing the effect of exchange between two species A and B. In the absence of exchange ($k_{\text{ex}} = 0 \text{ s}^{-1}$, bottom left), separate lines are seen for the two species and the widths of these lines are determined by the transverse relaxation rate constant; in this simulation their frequency separation is 160 Hz. It is assumed that the populations of A and B are equal, so the two lines have the same intensity. As the exchange rate constant increases the lines first broaden and then merge into one, as shown in the left-hand series of spectra. A further increase in the exchange rate constant causes the merged line to narrow, as shown in the right-hand series of spectra. In the limit that the exchange rate constant is very much larger than the frequency separation the exchange process has no effect on the linewidth. Note that scale expansions have been used for some of the spectra on the left.

the populations of A and B, and since we have assumed these to be equal, the merged line lies halfway between the two separate resonances.

An increase in the linewidth as a result of an exchange process is called *exchange broadening*. If we start out from the slow exchange limit, increasing the rate constant for the exchange process increases the amount of exchange broadening. However, after coalescence further increases in the rate constant cause a reduction in the linewidth. This phenomena is called *exchange narrowing*.

Exchange processes from the point of view of single spins

So far we have described the overall spectra which arise from species undergoing chemical exchange. Necessarily, these spectra are observed on macroscopic samples containing vast numbers of spins, so what we see is the average behaviour of all of these spins. What we are now going to explore is the microscopic behaviour, at the level of an individual spin, which explains the spectra we observe from the whole sample.

As before, we will consider the simple equilibrium between two species A and B, but this time focus in on the behaviour of a particular nuclear spin. Let us assume that this nucleus finds itself in an A molecule. As a result, the magnetic moment of this spin will precess at the Larmor frequency $\omega_{0,A}$ appropriate for this environment, and this tiny contribution will be added to that made by all of the other spins so as to give the FID. At some point, the A molecule may undergo a chemical transformation to B. All this involves is a rearrangement of the nuclei, so our spin now finds itself in a B molecule and the magnetic moment now precesses at the Larmor frequency

A full discussion of the theory used to simulate the spectra in Fig. 9.29 is given in J. Cavanagh, W. J. Fairbrother, A. G. Palmer III, M. Rance and N. J. Skelton, *Protein NMR Spectroscopy* (2nd edition, Academic Press, 2007), pages 391 to 400.

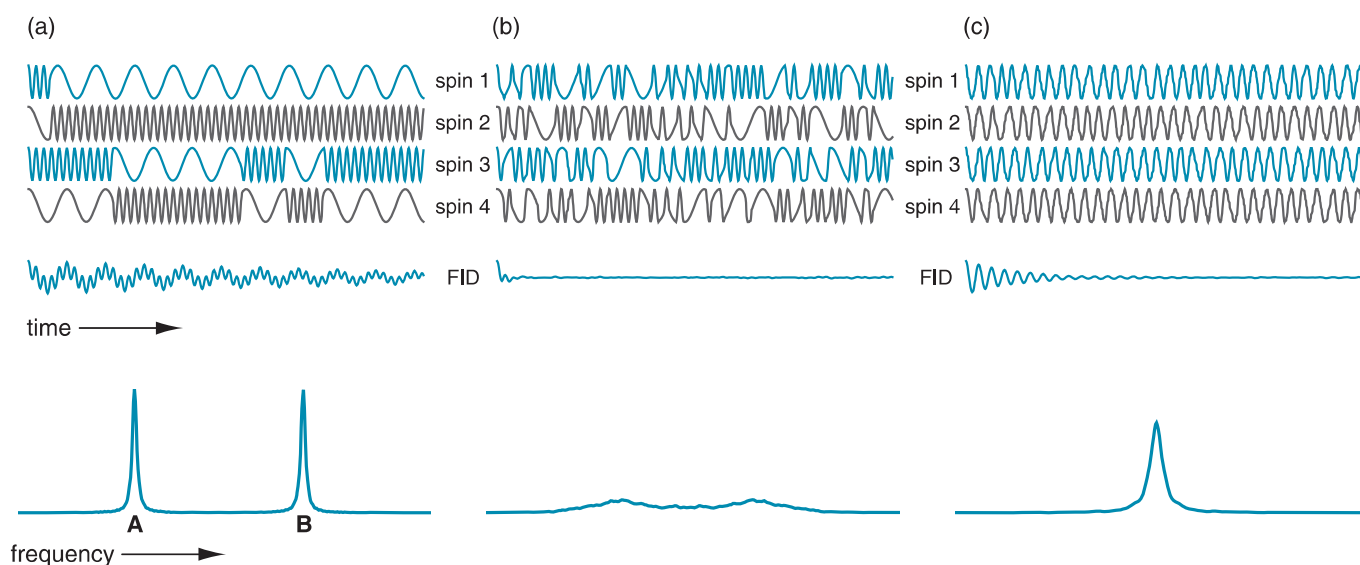


Fig. 9.30 Simulations showing how the behaviour of individual spins can account for the form of the spectra seen for the $A \rightleftharpoons B$ exchange system. At the top of (a), (b) and (c) are shown the contributions from four typical spins. Underneath, labelled FID, is the sum of the contributions from 1000 spins: this is a reasonable approximation to the FID which would be observed from a macroscopic sample. Beneath each FID is shown the corresponding spectrum. In (a) the number of transformations between A and B are rather few so distinct periods of oscillation at the two Larmor frequencies can be seen. The result is two clear lines in the spectrum, corresponding to the slow exchange limit. In (b) the transformations are more frequent, and as a result the contributions of the individual spins soon cancel one another out, leading to a quickly decaying FID and broad lines in the spectrum. In (c) the transformations are so frequent that each spin appears to be evolving at the average frequency of A and B. The individual contributions do not cancel one another as quickly as in (b), leading to a sharper line.

$\omega_{0,B}$, thus making a contribution to the FID at this frequency. At some time later, the molecule may transform back into A once again carrying the spin with it; now the spin makes a contribution at $\omega_{0,A}$.

Therefore, at the level of a single spin all that happens is that, as the molecule transforms back and forth between A and B, the Larmor frequency of the spin hops back and forth between $\omega_{0,A}$ and $\omega_{0,B}$. For a given spin, the time between transformations is essentially random, but the faster the overall rate of the chemical process the shorter the average time between transformations.

Figure 9.30 visualizes this process by showing the contribution from individual spins, as well as the sum of the contributions from many spins. Consider first the left-hand side of the diagram, (a). Shown at the top are the contributions to the transverse magnetization made by four individual spins. The Larmor frequencies in the two environments have been chosen to be sufficiently different that in the diagram they can be discerned by eye.

The abrupt change in the frequency of the oscillation comes about when the molecule transforms from A to B, or vice versa. Over the timescale of the diagram, only a few transformations take place, and as a result we can see extended periods during which an individual spin is precessing at one

of the two Larmor frequencies. Note that the changes take place at random times and are not correlated between the different spins.

In order to work out what the spectrum looks like from a macroscopic sample we need to add together the contributions from a very large number of spins. In the figure, this has been done for just 1000 spins (in order to make the simulation feasible), giving the time-domain function marked FID. The Fourier transformation of this gives the spectrum shown.

Not surprisingly, for (a) the time-domain signal clearly shows the presence of two frequencies, as is confirmed by the two peaks in the spectrum. What is going on here is that each individual spin spends long enough precessing at one or other of the two Larmor frequencies that distinct contributions to the overall time-domain signal are seen at these two frequencies.

Now consider Fig. 9.30 (b), in which the transformations between **A** and **B** are more frequent than they were in (a). However, we can still just about discern periods of oscillation at the two different frequencies. Adding up 1000 of these contributions from individual spins gives the time-domain signal labelled FID. What has happened here is that the frequent jumps from one frequency to another have led the contributions from the individual spins to cancel one another out, leading to a rapidly decaying FID and hence broad lines in the spectrum.

Finally consider part (c). Here the $\mathbf{A} \rightleftharpoons \mathbf{B}$ transformations occur so frequently that we cannot see periods of oscillation at either of the two frequencies. Rather, what we have for each spin is an oscillation at a single frequency, intermediate between the two seen in (a), arising from the fact that the frequency at which the spin is evolving is alternating rapidly between $\omega_{0,A}$ and $\omega_{0,B}$. Admittedly the oscillation is not a pure cosine wave, but its period is nevertheless clear. Adding up contributions from 1000 such spins gives some cancellation at later times, but nothing like the extent seen in (b); in addition, it is clear that there is a single frequency present in the FID. The resulting spectrum has a line broader than in (a), but much narrower than in (b).

The interpretation of these results is that Fig. 9.30 (a) corresponds to slow exchange, (b) corresponds to a situation approaching coalescence in which there is considerable line broadening, and (c) corresponds to a regime in which, although the two lines have collapsed to give one, there is still a significant exchange contribution to the linewidth. The spectra are analogous to those shown in Fig. 9.29 on page 287, although they were generated by a completely different method. We can therefore explain the spectra arising from chemically exchanging systems in terms of the behaviour of individual spins.

The conditions for slow and fast exchange

It has already been mentioned that whether or not we are in the fast or slow exchange region depends on the comparison between the rate constant for exchange and the *difference* in the Larmor frequencies between the two sites. However, Fig. 9.30 perhaps does not quite convey the impression that it is this frequency difference which is the important parameter.

To explore this further consider a ‘thought’ experiment in which we are required to tell the difference between a wave oscillating at 10.0 Hz and

In the simulation used to create Fig. 9.30 an extra decay has been imposed on the FID in order to make it more realistic. This rate of this decay gives the linewidth in the absence of chemical exchange effects, and the same decay has been used for all parts of the figure.

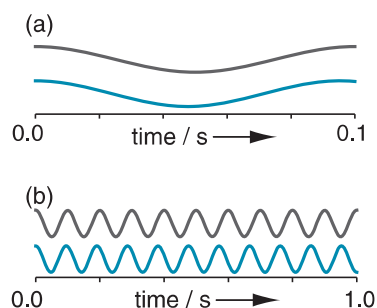


Fig. 9.31 Illustration of the time needed to differentiate between two cosine waves of different frequencies. In both (a) and (b) two cosine waves with frequencies 10.0 Hz and 10.5 Hz are plotted in black and blue, respectively. If the cosine waves are observed for 0.1 s, as in (a), it is very hard to see that they have different frequencies. However, if the observation is made for 1.0 s, as in (b), it is immediately obvious that the frequencies of the two waves are different.

one oscillating at 10.5 Hz. The question is, for how long would we need to observe these oscillations before we could be certain which was which?

The answer to this question is illustrated in Fig. 9.31. In (a) we see a plot of the two waves over a time interval of 0.1 s. Each oscillation barely completes one cycle, and we would have to look very hard to spot which is which. If there were any appreciable noise present, we surely would not be able to discern any difference between the two curves.

In contrast, if the oscillations are observed for a longer time, such as the 1.0 s shown in (b), it is quite easy to spot that the two oscillations have different frequencies since we can see that they are getting out of step with one another. By the time one second has elapsed the blue trace has completed almost half a cycle more than the black trace, so the former must be the higher frequency oscillation. Put more formally, by the end of the time period in (b) there is a significant phase difference between the two oscillations, whereas in (a) there is not. It is this difference in phase which makes it possible to tell the two waves apart.

The phase difference which develops over a time τ is $2\pi\Delta \times \tau$, where Δ is the difference in frequency (in Hz) between the two sites. For this phase difference to be significant τ must be comparable with or greater than $1/\Delta$. In other words, to be able to differentiate two frequencies we need to observe for a time which is greater than the reciprocal of the difference between the two frequencies.

In the case of chemical exchange characterized by a first-order rate constant k_{ex} , the average time before **A** is converted to **B** (or vice versa) is of the order of $1/k_{\text{ex}}$: this is often called the lifetime of the reaction, τ_{ex} . We are only able to observe the oscillations at a particular frequency for this time, because on average this is how long there is before the frequency changes.

If the rate of the process is such that $\tau_{\text{ex}} \gg 1/\Delta_{\text{AB}}$, where Δ_{AB} is the difference in frequency between the two sites, then it will be possible to distinguish between these two frequencies and so we expect to see two lines in the spectrum: this is the slow exchange limit. Replacing τ_{ex} with $1/k_{\text{ex}}$ and rearranging gives the condition in its familiar form:

$$\text{slow exchange } k_{\text{ex}} \ll \Delta_{\text{AB}}.$$

Conversely, if $\tau_{\text{ex}} \ll 1/\Delta_{\text{AB}}$ the time is insufficient to distinguish between the two frequencies, and so we expect there to be one line in the spectrum: this is the fast exchange limit. As before, substitution for τ_{ex} and rearrangement gives the familiar result

$$\text{fast exchange } k_{\text{ex}} \gg \Delta_{\text{AB}}.$$

9.8.2 The secular contribution to transverse relaxation

We can now draw on this discussion of the effects of chemical exchange to understand the origin of the secular contribution to transverse relaxation. Imagine a sample consisting of many spins, each of which experiences a local field due, for example, to interactions with nearby spins. The z -component of the local field at a particular spin will determine the Larmor frequency of that spin. Since we expect there to be a distribution of local fields, there will be a distribution of Larmor frequencies across the sample.

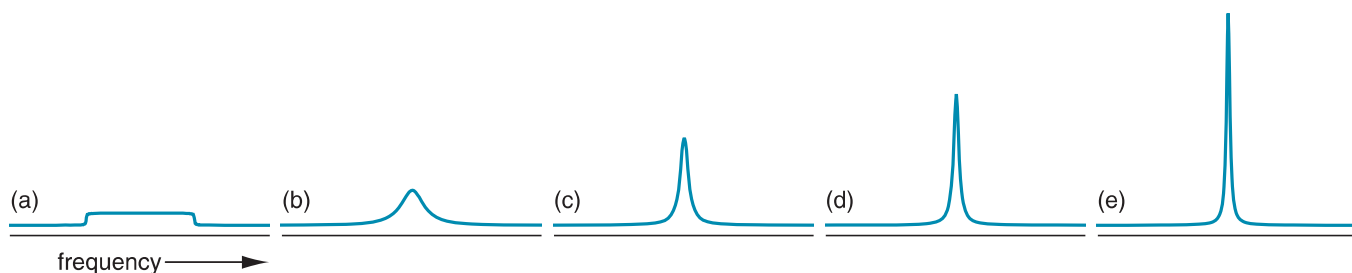


Fig. 9.32 Simulations showing the effect on the lineshape of exchange between spins with a continuous range of Larmor frequencies. In (a) is depicted the situation in which there is no exchange between the spins. As a result the line is broad, reflecting the range of Larmor frequencies present. Spectra (b)–(e) show the effect of increasing the exchange rate between the spins. The faster the exchange, the narrower the line becomes in a way analogous to two-site exchange after coalescence. In (e), although the line is much narrower than in (a), there is still a contribution arising from the original spread of Larmor frequencies shown in (a): this is the origin of the secular part of transverse relaxation.

First consider the situation in which the local fields do not change with time. Each spin will make a contribution to the FID at its own Larmor frequency, and so the distribution of Larmor frequencies across the sample will result in the spectrum consisting of a very broad line. This is shown schematically in Fig. 9.32 (a). The width of the line depends on the spread of values of the local field. For example, in the case where the fields are caused by dipolar interaction between two protons, the line could easily be 100 kHz wide.

Of course, in a liquid the local fields are not constant but, as described in section 9.3 on page 251, change rather rapidly as a result of the thermal motion (rotational diffusion) in the sample. An individual spin therefore experiences a local field which, on average, changes after a time of the order of the correlation time τ_c . The change in the component of the local field along the z -axis results in a change in the Larmor frequency. This is entirely analogous to chemical exchange in which the transformation of the molecule results in a change in the Larmor frequency experienced by a spin. The only difference is that rather than the spin experiencing two Larmor frequencies it can experience a whole range of Larmor frequencies.

In the previous section it was noted that the rate of an exchange process could be characterized by a lifetime τ_{ex} which is simply $1/k_{\text{ex}}$. In the case of rotational diffusion we can identify τ_{ex} with the correlation time τ_c , since both characterize the timescale of the process. For a non-viscous liquid τ_c might be around 100 ps, which corresponds to a ‘rate constant’ of $1/(100 \times 10^{-12}) = 10^{10} \text{ s}^{-1}$ – a very large value. The rate constant is very much larger than the spread of Larmor frequencies due to the local fields, we are clearly in the exchange narrowing region. Therefore we expect the rapid motion to result in significant narrowing of the broad line. The faster the motion, that is the shorter the correlation time, the greater the narrowing effect, as shown in Fig. 9.32 (b)–(e).

For the simple $A \rightleftharpoons B$ exchange it can be shown that, after coalescence, the exchange contribution to the linewidth is

$$\frac{\pi \Delta_{AB}^2}{2k_{ex}}.$$

We can use this expression to make a very rough estimate of the width to which our initial broad line is reduced by the motion. Replacing k_{ex} by $1/\tau_c$ and Δ_{AB} by W , the width of the distribution of Larmor frequencies (in Hz), we obtain

$$\text{width of narrowed line} \approx W^2 \times \tau_c. \quad (9.30)$$

This estimate is so rough that we have simply dropped the factor of $\pi/2$ as being irrelevant.

Putting $W = 100$ kHz and $\tau_c = 100$ ps, we obtain a value of 1 Hz for the width of the line. Clearly the motion has had a dramatic narrowing effect, taking the linewidth from 100 kHz in the absence of motion to just 1 Hz when motion is taken into account. However, for high-resolution NMR of liquids, 1 Hz is still a significant contribution to the linewidth. Note that the linewidth scales directly with the correlation time. Therefore if $\tau_c = 10$ ns, a value typical for a medium-size protein, the linewidth becomes 100 Hz, which is very significant.

In section 9.3.2 on page 256 we introduced the reduced spectral density $j(\omega)$ which gives a measure of the amount of motion at frequency ω . For the simplest kind of rotational diffusion this spectral density is given by:

$$j(\omega) = \frac{2\tau_c}{1 + \omega^2\tau_c^2}.$$

The spectral density at zero frequency is therefore $2\tau_c$. The rough estimate given in Eq. 9.30 can therefore be written as

$$\text{width of narrowed line} \approx \frac{1}{2}W^2 \times j(0),$$

where $j(0)$ is the reduced spectral density at zero frequency.

We can now see that the origin of the secular part of transverse relaxation lies in the distribution of local fields along the z -axis, and hence in the resulting distribution of Larmor frequencies. However, rapid motion in the liquid means that the observed line is very much narrower than the original distribution of Larmor frequencies. By drawing an analogy with chemical exchange we have been able to predict that the width of the narrowed line will be proportional to τ_c i.e. the value of the reduced spectral density at zero frequency, $j(0)$.

This brings us back to the discussion at the start of section 9.1.4 on page 247 in which the two contributions to transverse relaxation were described. The non-secular contribution involves reorientation of the magnetic moments of individual spins and so is identical to the process which leads to longitudinal relaxation. It therefore depends on the spectral density at the Larmor frequency. In contrast, the secular contribution arises from the motionally narrowed distribution of Larmor frequencies arising from the z -components of the local fields. The motionally narrowed linewidth, and hence the contribution to transverse relaxation, depends on the spectral density at zero frequency.

Terminology

The terms secular and non-secular come from quantum mechanics. In this theory, a secular perturbation is one which changes the energy, but not the wavefunction, whereas a non-secular perturbation changes both. An oscillating transverse magnetic field causes an individual magnetic moment to change orientation, and so changes the wavefunction – it is thus a non-secular effect. A field along the z -axis will alter the rate of precession of the magnetic moment, which is the same as altering the energy, but does not alter the orientation with respect to the z -axis – this is a secular effect.

9.8.3 Effect of transverse relaxation

In section 9.4.2 on page 260 we showed that, for a sample consisting of single spins, the relaxation behaviour of the z -magnetization is given by (Eq. 9.12 on page 262)

$$\frac{dM_z(t)}{dt} = -R_z [M_z(t) - M_z^0].$$

This equation predicts that relaxation will drive the z -magnetization to its equilibrium value of M_z^0 .

The transverse magnetization, for example the x -magnetization, obeys the following differential equation:

$$\frac{dM_x(t)}{dt} = -R_{xy} M_x(t), \quad (9.31)$$

where R_{xy} is the rate constant for transverse relaxation and, for simplicity, it has been assumed that the offset is zero. There is a similar equation for the y -magnetization, with the same rate constant. We will now show that this equation predicts that the x -magnetization decays exponentially to its equilibrium value of zero.

All that we have to do is to integrate Eq. 9.31 using the same approach as in section 9.5 on page 263:

$$\begin{aligned} \frac{dM_x(t)}{dt} &= -R_{xy} M_x(t) \\ \frac{dM_x(t)}{M_x(t)} &= -R_{xy} dt \\ \int \frac{dM_x(t)}{M_x(t)} &= \int -R_{xy} dt \\ \ln M_x(t) &= -R_{xy} t + \text{const.} \end{aligned}$$

We can find a value for the constant by taking the value of the x -magnetization at time $t = 0$ to be $M_x(0)$. Substituting this into the last line shows that the constant of integration is $\ln M_x(0)$, so the solution to the differential equation is

$$\ln M_x(t) = -R_{xy} t + \ln M_x(0).$$

This can be rearranged as follows:

$$\begin{aligned}
 \ln M_x(t) - \ln M_x(0) &= -R_{xy} t \\
 \ln \left(\frac{M_x(t)}{M_x(0)} \right) &= -R_{xy} t \\
 \frac{M_x(t)}{M_x(0)} &= \exp(-R_{xy} t) \\
 M_x(t) &= M_x(0) \exp(-R_{xy} t).
 \end{aligned}$$

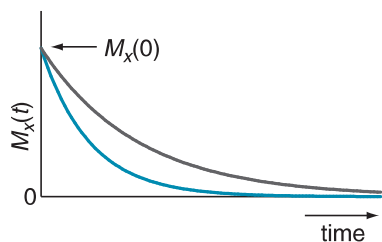


Fig. 9.33 Transverse magnetization decays exponentially at a rate determined by the transverse relaxation rate constant, R_{xy} . Curves are shown for two values of the rate constant, with the blue curve having twice the rate constant of the dark grey curve.

What the last line says is that the x -magnetization decays from its initial value to the equilibrium value of zero at a rate determined by the rate constant R_{xy} ; typical examples of such a decay are shown in Fig. 9.33. The y -magnetization behaves in exactly the same way.

The reciprocal of the rate constant R_{xy} is the time constant for the decay of transverse magnetization. Usually this time constant is denoted T_2 , where $T_2 = 1/R_{xy}$. The use of this symbol is so widespread that transverse relaxation is often called ‘ T_2 relaxation’.

If the offset is not zero then, in addition to decaying due to transverse relaxation, the x - and y -components of the magnetization are interconverted due to their precession in the transverse plane. If we are only interested in the relaxation part of this process, we can eliminate the effect of precession by defining M_{xy} as

$$M_{xy} = \sqrt{M_x^2 + M_y^2}.$$

The value of M_{xy} does not change as a result of precession, so its time dependence is just determined by relaxation:

$$\frac{dM_{xy}(t)}{dt} = -R_{xy} M_{xy}(t).$$

9.8.4 Relaxation by random fields

One of the simplest problems we can deal with in the theory of relaxation is to imagine that our spin experiences a random local field. We do not ask where the field comes from, but just accept that it is there.

To make things as simple as possible we will also assume that the mean square of the local field is the same in the x -, y - and z -directions:

$$\overline{B_{\text{loc},x}^2} = \overline{B_{\text{loc}}^2}, \quad \overline{B_{\text{loc},y}^2} = \overline{B_{\text{loc}}^2}, \quad \overline{B_{\text{loc},z}^2} = \overline{B_{\text{loc}}^2}.$$

With these assumptions, it can be shown that the rate constant for longitudinal relaxation is given by:

$$R_z = \gamma^2 \overline{B_{\text{loc}}^2} j(\omega_0), \quad (9.32)$$

where γ is the gyromagnetic ratio and ω_0 is the Larmor frequency (we are not able to go into the details of the theory used to compute these rate constants). As expected, the rate of longitudinal relaxation depends on the spectral density at the Larmor frequency.

The rate constant for transverse relaxation, R_{xy} , can be shown to be given by

$$R_{xy} = \underbrace{\frac{1}{2}\gamma^2 \overline{B_{\text{loc}}^2} j(0)}_{\text{secular}} + \underbrace{\frac{1}{2}\gamma^2 \overline{B_{\text{loc}}^2} j(\omega_0)}_{\text{non-secular}}.$$

As expected, the rate constant has a secular part, depending on $j(0)$, and a non-secular part, depending on $j(\omega_0)$. For the secular part the term $\gamma^2 \overline{B_{\text{loc}}^2}$ is analogous to the square of the width of the distribution of Larmor frequencies caused by the local fields, as described in section 9.8.2 on page 290.

The new feature which these calculations reveal is that the non-secular part is precisely *half* the value of the longitudinal rate constant given in Eq. 9.32 on the facing page:

$$R_{xy} = \underbrace{\frac{1}{2}\gamma^2 \overline{B_{\text{loc}}^2} j(0)}_{\text{secular}} + \underbrace{\frac{1}{2}R_z}_{\text{non-secular}}.$$

It turns out that for *any* relaxation mechanism it is always true that there is a contribution to the transverse relaxation rate constant which is equal to one-half of the longitudinal rate constant. This factor of one-half cannot be a coincidence, so where does it come from? One argument goes along the following lines.

Imagine that a particular spin experiences a local field, the transverse component of which is aligned along the x -axis and oscillating at the Larmor frequency. The z -component of the magnetic moment of the spin will thus be rotated, as will its y -component. However, the x -component is unaffected, as it is parallel with the local field. Thus, the z -component of the moment is altered, leading to longitudinal relaxation, but only *one* of the transverse components is altered. We interpret this by saying that the field is half as effective at causing transverse relaxation as it is at causing longitudinal relaxation, as only one of the two transverse components of the moment is affected.

Relaxation rates in the two motional limits

It is interesting to see how the longitudinal and transverse relaxation rate constants compare in the fast and slow motion limits. Recall that in the fast motion limit $j(\omega_0) = 2\tau_c$ and, independent of the motional regime, $j(0) = 2\tau_c$. The two rate constants are thus

$$\begin{aligned} \text{fast motion: } R_z &= \gamma^2 \overline{B_{\text{loc}}^2} j(\omega_0) \\ &= 2\gamma^2 \overline{B_{\text{loc}}^2} \tau_c, \end{aligned}$$

and

$$\begin{aligned} \text{fast motion: } R_{xy} &= \frac{1}{2}\gamma^2 \overline{B_{\text{loc}}^2} j(0) + \frac{1}{2}\gamma^2 \overline{B_{\text{loc}}^2} j(\omega_0) \\ &= \gamma^2 \overline{B_{\text{loc}}^2} \tau_c + \gamma^2 \overline{B_{\text{loc}}^2} \tau_c \\ &= 2\gamma^2 \overline{B_{\text{loc}}^2} \tau_c. \end{aligned}$$

Therefore, in the fast motion limit the two rate constants are equal, which is a consequence of the spectral density being independent of frequency.

In the slow motion limit $j(0)$ is still $2\tau_c$, and as described in section 9.3.3 on page 257, $j(\omega_0)$ is given by

$$j(\omega_0) = \frac{2}{\omega_0^2 \tau_c}.$$

So, the rate constant for longitudinal relaxation becomes

$$\begin{aligned} \text{slow motion: } R_z &= \gamma^2 \overline{B_{\text{loc}}^2} j(\omega_0) \\ &= \frac{2\gamma^2 \overline{B_{\text{loc}}^2}}{\omega_0^2 \tau_c}. \end{aligned}$$

Note that this rate constant *decreases* as the correlation time increases.

In contrast, the rate constant for transverse relaxation goes on increasing as τ_c increases on account of the secular term:

$$\begin{aligned} \text{slow motion: } R_{xy} &= \frac{1}{2} \gamma^2 \overline{B_{\text{loc}}^2} j(0) + \frac{1}{2} \gamma^2 \overline{B_{\text{loc}}^2} j(\omega_0) \\ &= \gamma^2 \overline{B_{\text{loc}}^2} \tau_c. \end{aligned}$$

On the second line we have simply set $j(\omega_0)$ to zero, as it will be negligible compared with $j(0)$.

Figure 9.34 shows a plot of the two rate constants as a function of the correlation time. The fast motion limit, in which the two rate constants are equal, is seen when τ_c is small. As the correlation time increases, the longitudinal relaxation rate constant eventually reaches a maximum. It was shown in section 9.3.2 on page 256 that $j(\omega_0)$ is a maximum when $\omega_0 \tau_c = 1$, so this is the condition for R_z being a maximum. As the correlation time increases further, R_z falls off steadily. In contrast, the transverse relaxation rate constant goes on increasing with τ_c .

The practical consequence of these observations is that for large molecules, which tumble slowly and so have long correlation times, transverse magnetization decays away to zero much more quickly than the z -magnetization recovers to equilibrium.

9.8.5 Transverse dipolar relaxation of two spins

For two spins relaxing by the dipolar interaction the following relaxation behaviour is found:

$$\frac{dI_{1x}}{dt} = -R_{xy}^{(1)} I_{1x} \quad \frac{dI_{2x}}{dt} = -R_{xy}^{(2)} I_{2x},$$

where

$$R_{xy}^{(1)} = b^2 \left[\underbrace{\frac{1}{10} j(0)}_{\text{secular}} + \underbrace{\frac{3}{20} j(\omega_{0,2}) + \frac{3}{40} j(\omega_{0,1}) + \frac{3}{20} j(\omega_{0,1} + \omega_{0,2}) + \frac{1}{40} j(\omega_{0,1} - \omega_{0,2})}_{\text{non-secular} = R_z^{(1)}/2} \right],$$

and

$$R_{xy}^{(2)} = b^2 \left[\underbrace{\frac{1}{10} j(0)}_{\text{secular}} + \underbrace{\frac{3}{20} j(\omega_{0,1}) + \frac{3}{40} j(\omega_{0,2}) + \frac{3}{20} j(\omega_{0,1} + \omega_{0,2}) + \frac{1}{40} j(\omega_{0,1} - \omega_{0,2})}_{\text{non-secular} = R_z^{(2)}/2} \right].$$

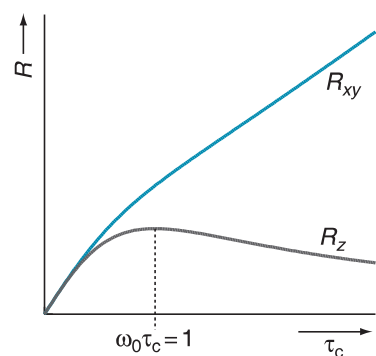


Fig. 9.34 Plot of the rate constants for longitudinal (R_z) and transverse (R_{xy}) relaxation, caused by random fields, as a function of the correlation time, τ_c . In the fast motion limit ($\omega_0 \tau_c \ll 1$), the two rate constants are equal, and increase with τ_c . However, as the correlation time increases further the rate constant for longitudinal relaxation reaches a maximum and then falls off. In contrast, the rate constant for transverse relaxation goes on increasing indefinitely, eventually becoming a linear function of τ_c .

As before, the non-secular contribution is equal to one-half of the corresponding longitudinal rate constant, expressions for which were given in Eq. 9.20 on page 272.

Of the two terms which have been labelled secular, the first needs no comment since, as in the case of random fields, it depends on the spectral density at zero frequency. In the expression for $R_{xy}^{(1)}$, that is the transverse relaxation rate constant for *spin one*, the second secular term depends on the spectral density at the Larmor frequency of *spin two*, $\omega_{0,2}$. Our interpretation of this term is as follows.

Spin one sees a magnetic field due to spin two and, as we have seen, this field depends on the distance between the two spins and the orientation of the vector joining them. However, the field at spin one also depends on the spin state of spin two i.e. whether it is up or down. Flipping spin two causes the local field experienced by spin one to change, thus making a contribution to the spread of local fields. The rate of flipping of spin two depends on the spectral density at the Larmor frequency of this spin, as it is motion at this frequency which is needed to flip the spin.

As with relaxation by random fields, in the fast motion limit $R_{xy} = R_z$, and in addition R_{xy} is the same for both spins involved in the dipolar interaction.

9.8.6 Transverse cross relaxation: ROESY

For two spins undergoing dipolar relaxation, we saw that the behaviour of the z -magnetization of spin one depends on the deviation of spin two from equilibrium, and vice versa. This effect appears in the Solomon equations via the cross-relaxation term depending on σ_{12} :

$$\begin{aligned}\frac{dI_{1z}}{dt} &= -R_z^{(1)}(I_{1z} - I_{1z}^0) - \sigma_{12}(I_{2z} - I_{2z}^0) \\ \frac{dI_{2z}}{dt} &= -\sigma_{12}(I_{1z} - I_{1z}^0) - R_z^{(2)}(I_{2z} - I_{2z}^0).\end{aligned}$$

However, for transverse relaxation, no such cross term is seen between the two transverse terms:

$$\frac{dI_{1x}}{dt} = -R_{xy}^{(1)}I_{1x} \quad \frac{dI_{2x}}{dt} = -R_{xy}^{(2)}I_{2x}.$$

The question is, why not?

The answer to this question is that, in principle, there *is* such a cross-relaxation term between I_{1x} and I_{2x} – it is just that we cannot observe the effect of this term. To see why this is, imagine that spin one is on resonance and that its transverse magnetization is stationary along the x -axis. Spin two, however, does have an offset and so its transverse magnetization precesses around in the transverse plane.

Cross relaxation can cause some of the spin-two transverse magnetization to become spin-one transverse magnetization. This new contribution to spin one will appear in the direction in which the spin-two magnetization is pointing. A little time later, further cross relaxation generates some more spin-one transverse magnetization. As the spin-two magnetization has precessed around, the new contribution to spin one will be in a different